Structural Compatibility Modeling for Apartment-Level Tenant Allocation in Co-Living

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Abstract

(English Version)

Commercial co-living offers a housing product in which tenants do not simply rent a room, but also share everyday private space with previously unknown flatmates. Despite this, tenant allocation is often shaped more by availability and chance than by any structured assessment of interpersonal fit. This thesis develops a data-driven compatibility framework to support flatmate grouping and apartment recommendation in co-living environments.

The study follows a two-stage design. Stage 1 examines personality as a basis for interpersonal fit by constructing pairwise compatibility scores from Big Five trait profiles in the IPIP dataset. Rather than relying on similarity alone, the framework allows for trait-specific weighting and selective complementarity, and uses simulated group formation to explore how pairwise compatibility can be translated into apartment-level outcomes. Stage 2 extends this logic by combining real operational tenant and apartment data from a co-living operator with synthetically constructed lifestyle and compatibility-related variables to evaluate existing shared-living groups and support applicant-to-apartment recommendations under vacancy conditions.

Apartment-level compatibility is modeled by aggregating pairwise scores into group-level measures of average fit and lower-tail compatibility, providing a transparent basis for allocation support. An interactive applicant-facing prototype translates these outputs into ranked apartment recommendations, while a complementary operator-facing dashboard provides comparative views of apartment composition and compatibility patterns across the housing portfolio. Although empirical validation is limited by the lack of observed compatibility outcomes and the reliance on synthetic Stage 2 inputs, the results demonstrate the feasibility of a structured and interpretable compatibility framework for shared housing.

Keywords: Tenant compatibility; Co-living; Compatibility modeling; Apartment-level allocation; Pairwise compatibility; Group assignment; Recommendation systems; Decision-support systems.

Title: Structural Compatibility Modeling for Apartment-Level Tenant Allocation in Co-Living

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Resumo

(Versão Portuguesa)

O co-living comercial oferece um produto habitacional em que os inquilinos arrendam um quarto e partilham espaço privado do quotidiano com pessoas desconhecidas. Apesar disso, a alocação continua a ser guiada pela disponibilidade e pelo acaso, e não por uma avaliação estruturada da compatibilidade interpessoal. Esta dissertação desenvolve um quadro de compatibilidade orientado por dados para apoiar a formação de grupos de residentes e a recomendação de apartamentos em contextos de co-living.

O estudo segue uma estrutura em duas etapas. A Etapa 1 examina a personalidade como base do ajustamento interpessoal, construindo pontuações de compatibilidade par a par a partir de perfis Big Five do conjunto IPIP. Em vez de se basear apenas na similaridade, o modelo permite ponderações por traço e complementaridade seletiva. A Etapa 2 alarga esta lógica, combinando dados operacionais reais de inquilinos e apartamentos com variáveis sintéticas de estilo de vida e compatibilidade, para avaliar grupos residenciais existentes e apoiar recomendações para apartamentos com vagas.

A compatibilidade ao nível do apartamento é modelada pela agregação de pontuações par a par em medidas de ajustamento médio e compatibilidade na cauda inferior. Um protótipo orientado para candidatos traduz estes resultados em recomendações hierarquizadas de apartamentos, enquanto um painel orientado para operadores oferece perspetivas comparativas sobre composição e padrões de compatibilidade no portefólio habitacional. Embora a validação empírica seja limitada pela ausência de resultados observados de compatibilidade e pela dependência de inputs sintéticos na Etapa 2, os resultados demonstram a viabilidade de um quadro interpretável para habitação partilhada.

Título: Modelação Estrutural da Compatibilidade para a Alocação de Inquilinos ao Nível do Apartamento em Co-Living

Palavras-chave: Compatibilidade de inquilinos; Co-living; Modelação de compatibilidade; Alocação ao nível do apartamento; Compatibilidade par a par; Formação de grupos; Sistemas de recomendação; Sistemas de apoio à decisão.

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Declaration on AI Use

Generative AI tools were used in a limited and supportive capacity during the preparation of this thesis. Their use was restricted primarily to language refinement, structural suggestions, formatting support, occasional coding assistance, and selective support in the preparation of synthetic example data. They were not used to replace the author's independent research design, analysis, interpretation, or academic judgement.

All substantive methodological choices, data preparation decisions, modeling steps, analysis, and conclusions were developed and validated by the author. Any AI-assisted output was critically reviewed, revised where necessary, and only incorporated after independent verification. Responsibility for the content, accuracy, and academic integrity of this thesis remains solely with the author.

Contents

1 Introduction **9**

1.1 Background and Significance 9

1.2 Motivation 9

1.3 Research Question 10

1.4 Research Objectives and Scope 11

1.4.1 Objectives 11

1.4.2 Scope 11

1.5 Research Gaps and Contribution 12

1.5.1 Research Gaps 12

1.5.2 Gaps to be Filled 12

2 Literature Review **13**

2.1 Theoretical Foundations of Interpersonal Compatibility 13

2.2 Compatibility in Shared Living and Co-Living Contexts 13

2.3 Personality as a Dimensional Trait Space 14

2.4 Quantitative Formalizations of Interpersonal Similarity 14

2.5 From Dyadic Compatibility to Group-Level Structure 15

2.6 Analytical Approaches to Matching and Assignment 16

3 Rationale and Problem Statement **17**

3.1 Rationale for the Current Study 17

3.2 Justification for the Study 17

3.3 Broader Implications 17

3.4 Problem Definition 18

3.5 Significance of the Study 18

4 Research Methodology **19**

4.1 Research Design 19

4.2 Data Sources 19

4.2.1 Secondary Data: Personality Dataset 19

4.2.2 Co-living Dataset and Synthetic Survey Construction 20

4.3 Data Preparation 20

4.3.1 Stage 1 – Personality Dataset 20

4.3.2 Stage 2 – Operational and Synthetic Co-living Data 20

4.4 Conceptual Compatibility Framework 22

4.5 Computational Design and Subsampling Strategy 23

5 Model Development **24**

5.1 Stage 1 – Personality-Based Compatibility 24

5.1.1 Trait Representation 24

5.1.2 Exploratory Distance/Similarity Metrics and Structural Inspection . . . 24

5.1.3 Compatibility Function Formulation 25

5.1.4 Pairwise Compatibility Matrix Construction 26

5.1.5 From Pairwise Compatibility to Group Assignment 26

5.2 Stage 2 – Extended Compatibility Framework 27

5.2.1 Feature Groups and Model Inputs 27

5.2.2 Personality Compatibility Layer 27

5.2.3 Lifestyle Compatibility Layer 28

5.2.4 Applicant-Personalized Combined Compatibility Layer 28

5.2.5 Vacancy-Level Structural Scoring 28

5.2.6 Context Flags and Interpretive Information 29

5.2.7 Recommendation and Matching Logic 29

6 Results and Evaluation **31**

6.1 Stage 1 – Personality-Based Compatibility 31

6.1.1 Exploration of Personality and Distance Structure 31

6.1.2 Compatibility Score Distribution and Sensitivity 31

6.1.3 Group Formation and Assignment Simulation 32

6.2 Stage 2 – Extended Compatibility Framework 32

6.2.1 Extended Feature Space Overview 32

6.2.2 Stage 2 Recommendation Setting 33

6.2.3 Applicant-Facing Recommendation Output 33

6.2.4 Operator Dashboard and Portfolio Monitoring 34

7 Challenges and Limitations **36**

7.1 Data Limitations 36

7.2 Methodological Constraints 37

7.3 Summary 37

8 Contributions **39**

8.1 Theoretical Contributions 39

8.2 Methodological Contributions 39

8.3 Practical Contributions 40

8.4 Educational and Curricular Relevance 40

9 Recommendations and Future Work **41**

9.1 Recommendations for Co-Living Operators 41

9.2 Model Enhancements 42

9.3 Future Research Directions 42

9.4 Concluding Remarks 43

References **44**

Appendix **46**

A.1 Stage 1: Descriptive Exploration 46

A.2 Stage 1: Distance Metrics and Compatibility Structure Scores 48

A.3 Stage 1: Sensitivity and Assignment Comparison 51

A.4 Stage 2: Descriptive Extensions 53

A.5 Tenant Compatibility Survey 58

1 Introduction

This chapter introduces the research topic, motivation, objectives, and research questions. It examines tenant compatibility in co-living through a data-driven framework and outlines the study's scope and contributions.

1.1 Background and Significance

The rather novel commercial co-living sector has grown steadily and is projected to reach a global market value of USD 16.05 billion by 2030, up from an estimated USD 7.82 billion in 2024 (Grand View Research, 2025). In response to this growth, many operators have refined their offerings around clearly defined tenant segments and increasingly provide shared amenities such as community spaces, fitness areas, and leisure facilities. While private living spaces are smaller in size compared to classical private flat-renting, shared amenities offer a community-spirited physical environment which has matured substantially from a product and service perspective.

Attributes that once distinguished leading co-living operators, including streamlined processes, curated interiors, and reliable high-speed internet, have gradually become baseline expectations across the market. As these physical and operational features become more standardized, true differentiation may only be achieved by tackling the difficult task of matching diverse tenant profiles within a shared apartment.

This sets the stage for tenant compatibility to become a central consideration in bettering the overall co-living product. Everyday fit may depend on aligned routines, household habits, and lifestyle preferences, while interpersonal fit - an indirect promise of offering a community space of similarly-situated individuals, may shape social comfort, friendship formation, and the overall harmony of the living environment. Operators can design and manage the physical setting. However, it remains far more difficult to anticipate how well residents will live together. This creates a strong case for structured, data-driven approaches that capture compatibility across multiple dimensions and support better tenant allocation, with potential benefits for satisfaction, retention, and well-being.

1.2 Motivation

The motivation for this study emerged from practical exposure to tenant allocation processes in shared housing. During an internship at a housing platform serving students and young professionals seeking mid-term accommodation in shared apartments across Europe, it became apparent that tenant selection was often guided by none or weakly structured criteria. In many cases, allocation decisions appeared largely opportunistic rather than systematically informed.

When prospective tenants asked who they would be sharing common areas with, the usual response was that no information could be disclosed for data protection reasons. While this position was legally justified, it also became an excuse for not dealing with the matching component and revealed how little attention was given to the interpersonal side of shared living. In effect, allocation often depended heavily on chance, despite the importance of comfort, trust, and social ease within the home environment.

This evident gap in the overall co-living product became even more apparent during subsequent work with a co-living operator in Berlin, with more than 500 rental units under management. Maintaining balanced apartment compositions was not treated as a primary operational concern, partially due to structural dispositions of the market such as shorter stays of around three months and high tenant fluctuation. Yet, it was clear that a well-functioning apartment could lose stability quickly when a poorly fitting tenant entered the group. Even so, no formal mechanism existed to assess compatibility characteristics or support more structured allocation decisions further than pointing out age or gender attributes as a loose anchor for matching criteria.

These experiences and observations motivated this research and the attempt to address tenant allocation in a more structural way. They pointed to a clear gap between the operational realities of co-living and the need for more deliberate tenant placement. The study therefore proceeds from the view that co-living operators can benefit from structured, data-driven approaches that represent relevant context-dependent compatibility characteristics and use them to support more balanced and transparent allocation decisions. Simultaneously, structuring tenant allocation through a lens of compatibility matching, at its core, is about achieving greater satisfaction for the individual tenant as well.

1.3 Research Question

Main question. How can tenant compatibility in co-living environments be conceptualized and operationalized through resident-relevant characteristics, and how can this be translated into a practical framework for apartment allocation?

Sub-questions.

SQ1. Which resident-related characteristics are most relevant for representing tenant compatibility in co-living environments?

SQ2. How can these characteristics be translated into a quantitative representation of individual tenants?

SQ3. How can compatibility between tenants and apartment groups be formally assessed within a quantitative modeling framework?

SQ4. How can these compatibility representations be integrated into a practical decision-support framework for tenant allocation in co-living apartments?

1.4 Research Objectives and Scope

1.4.1 Objectives

The objectives of this research are designed to address the main research question and its sub-questions by combining conceptual grounding with analytical framework development.

O1. Identify and structure the compatibility-relevant attributes of shared living environments through existing literature and industry-informed considerations.

O2. Develop a unified tenant representation that integrates psychological and lifestyle-related characteristics into a structured analytical profile.

O3. Formalize compatibility through similarity- and distance-based measures at both the tenant and apartment-group level.

O4. Design and operationalize a compatibility-based allocation framework that can support tenant placement decisions in co-living environments.

1.4.2 Scope

This research focuses on shared living environments operated by professional co-living providers in urban markets. The study is primarily concerned with young professionals and students in medium-term to long-term rental arrangements, typically upwards of three months.

The analysis centers on compatibility-relevant characteristics that can be collected and used in a practical and ethically defensible way, including personality-related traits, lifestyle preferences and household habits and selected demographic and temporal variables. In line with data protection considerations such as the GDPR, the framework places emphasis on behavioral and personality-based representations, while demographic information is incorporated more selectively as additional context flags. In the final allocation setting, such variables are not treated as the main drivers of compatibility but may function as constraints or transparency indicators, for example in relation to gender-specific apartments or notable age differences.

Rather than aiming to deliver a fully commercialized matchmaking solution, this thesis seeks to demonstrate the analytical feasibility of modeling tenant compatibility in a structured way and operationalizing it within a decision-support framework for shared apartment allocation and community management.

1.5 Research Gaps and Contribution

1.5.1 Research Gaps

Existing research has examined interpersonal compatibility in settings such as romantic relationships, workplace collaboration, and student roommate matching on the pairwise-level. However, comparatively little attention has been given to compatibility modeling in the context of professionally managed co-living environments. At the same time, tenant allocation in practice often remains experience-based, intuitive, or constrained by operational convenience rather than supported by explicit analytical frameworks.

In addition, although prior literature frequently discusses demographic, psychological, and lifestyle-related influences on compatibility, there is limited work on how such characteristics can be combined into a structured representation that supports both analytical assessment and practical allocation decisions in shared urban housing.

1.5.2 Gaps to be Filled

This research addresses these gaps by:

- adapting compatibility-related insights from adjacent research domains to the co-living context;
- developing a multi-dimensional tenant representation based on behavioral and personality-related characteristics;
- formalizing compatibility through similarity- and distance-based measures at both the interpersonal and at the aggregated apartment-group level; and
- translating these compatibility representations into a decision-support framework for structured tenant allocation.

Overall, the study contributes by linking research on interpersonal compatibility with the operational realities of co-living management. More specifically, it shows how compatibility can be represented, assessed, and incorporated into a structured allocation approach for shared living environments.

2 Literature Review

This chapter develops the conceptual and methodological foundation for the thesis. It reviews interpersonal compatibility as a relational construct in co-living and shared housing, positions personality as a multidimensional basis for representing individual differences, and discusses how similarity and compatibility can be formalized quantitatively and extended from dyads to groups. Finally, it considers analytical approaches to matching, assignment, and recommendation relevant to the study design.

2.1 Theoretical Foundations of Interpersonal Compatibility

Interpersonal compatibility is best understood as a relational form of fit that emerges between people rather than as an isolated characteristic of any one individual. In social and organizational research, compatibility has often been linked to attraction, cooperation, and the quality of ongoing interaction. A large meta-analysis by Montoya et al. (2008) showed that both actual similarity and perceived similarity are positively associated with interpersonal attraction, suggesting that some degree of congruence can facilitate affinity and ease of interaction. At the same time, compatibility cannot be reduced to uniform similarity alone. As argued by Tett and Murphy (2002), the social value of similarity and complementarity depends on the situation and on the functional role of the trait under consideration. More recent review work reiterates that attraction processes are shaped by personality, value congruence, and contextual interpretation rather than by simple resemblance on all dimensions (Wang, 2025).

For co-living environments, this broader interpretation is particularly important. Residents do not merely share space for short, task-specific encounters, but instead co-manage everyday routines, boundaries, and social expectations over time. Compatibility in this context therefore includes both practical fit and interpersonal fit. Similar routines, habits, and preferences may reduce everyday friction, while compatible personalities may support social ease, trust, and the possibility of friendship formation. In shared living contexts, compatibility therefore appears less as a matter of simple demographic resemblance or isolated behaviors and more as a multidimensional, context-dependent form of social fit.

2.2 Compatibility in Shared Living and Co-Living Contexts

Research on shared housing suggests that residents evaluate such environments through both economic and social-relational considerations. Reviews of shared housing among young adults describe it as shaped not only by financial pragmatism but also by preferred flatmates, conflict, and the delicate management of everyday domestic dynamics (Clark et al., 2018). This is partic-

ularly relevant in co-living, where accommodation is not only a physical housing arrangement but also a shared social environment.

Empirical work reinforces this interpretation. Cho et al. (2019) show that shared-housing experiences are associated with residential satisfaction and community attachment, suggesting that the value of shared living extends beyond cost savings alone. Similarly, Byun and Shon (2022) examine co-living as a lived residential setting and emphasize its relation to place attachment and the possibility of home. Other studies find that shared-housing characteristics shape housing satisfaction, and that housing satisfaction in turn relates to intentions to remain or recommend the housing arrangement to others (Baek and Kim, 2022).

2.3 Personality as a Dimensional Trait Space

Because interpersonal compatibility concerns how people are likely to experience, interpret, and respond to one another over time, the literature often approaches it through relatively stable individual differences. Personality is particularly relevant in this regard, as it provides a structured way to represent enduring tendencies in thought, affect, and behavior that can shape social interaction. The most widely used framework for capturing such variation is the Five-Factor Model, which organizes personality along the broad domains of extraversion, agreeableness, conscientiousness, emotional stability, and openness to experience (Widiger and Crego, 2019).

A dimensional personality framework is especially useful for compatibility research because it represents individuals as profiles across continuous trait dimensions rather than as fixed types. This preserves interpersonal variation, enables systematic comparison within a common trait space, and allows compatibility to be examined as a matter of relative resemblance or contrast across multiple characteristics. Research based on large-scale public-domain Big Five inventories further illustrates the robustness and practical applicability of dimensional personality measurement across empirical settings (Goldberg, 1992; Kajonius and Johnson, 2019). Taken together, this literature positions the Five-Factor Model as a useful basis for structured representations of interpersonal compatibility.

2.4 Quantitative Formalizations of Interpersonal Similarity

Although similarity is conceptually intuitive, its quantitative formalization is methodologically nontrivial. Early work by Gleser and Cronbach (1953) already demonstrated that different profile-comparison methods capture different aspects of resemblance between two multivariate profiles. This point remains highly relevant in applied compatibility modeling. Depending on the chosen metric, apparent similarity may reflect absolute closeness, relative profile shape, or

resemblance to a population norm rather than meaningful dyadic fit.

A major methodological concern in this literature is that raw profile similarity can be distorted by normativeness and distinctiveness. Furr (2008) argue that similarity should not be interpreted independently of whether two profiles are both close to the average profile or whether they share meaningful deviations from it. Related work shows that conclusions about similarity effects may vary depending on the operationalization that is used, and that simple difference scores are often insufficient for capturing the multidimensional logic of dyadic fit (Schönbrodt et al., 2018; Carlier et al., 2024). Taken together, these studies suggest that dyadic compatibility is sensitive to how similarity is defined and measured. Rather than treating resemblance as a single self-evident quantity, the literature indicates that different operationalizations capture different aspects of profile fit and may therefore lead to different substantive conclusions.

2.5 From Dyadic Compatibility to Group-Level Structure

Much of the literature conceptualizes interpersonal compatibility in dyadic terms. In co-living settings with multi-bedroom units where several residents share the same living environment or where one prospective tenant must be evaluated relative to an existing household, the practical challenge is to translate pairwise fit into an assessment of apartment-level compatibility. This raises the question of how a set of dyadic relations can be meaningfully aggregated into a group-level representation of fit.

Literature on coalition formation provides a useful conceptual bridge for addressing this problem. In fractional hedonic games, a person's utility within a coalition is defined as the average value that the person assigns to the other members of that coalition (Aziz et al., 2019). This framework is relevant because it shows how group quality can be derived from underlying pairwise evaluations rather than treated as an entirely separate construct. In settings such as shared apartments, this implies that apartment-level compatibility can be approached as an aggregation problem over the set of within-apartment interpersonal relations.

This literature also makes clear that group quality is not exhausted by a single criterion. Recent work on welfare in fractional hedonic games distinguishes between utilitarian and egalitarian objectives, thereby separating average coalition quality from the protection of the worst-off member (Hanaka et al., 2025). This distinction is especially pertinent in co-living environments, where an apartment may appear compatible on average while still containing one resident whose fit with the rest of the household is substantially weaker.

Related work on team formation in networks points in a similar direction. Lappas et al. (2009) show that once pairwise relations are known, the quality of a group can be evaluated in terms of the structure of connections among its members, for example through communication-cost

criteria. Gajewar and Sarma (2012) likewise emphasize the internal density of ties among selected members as an indicator of collaborative compatibility. All in all, these studies suggest that group-level fit is not simply a matter of geometric closeness in an original feature space, but can instead be understood as a structural property that emerges from the pattern of pairwise relations within the group. For shared apartments, this provides a useful literature-based rationale for treating apartment compatibility as an aggregation of dyadic fit rather than as a pure clustering problem.

2.6 Analytical Approaches to Matching and Assignment

Once compatibility has been formalized, the problem is no longer only how fit can be described, but also how it can be used in allocation. In the present context, this means deciding how applicant–apartment fit should be incorporated into the assignment of prospective tenants to apartments with limited capacity. One relevant body of literature here is matching theory, which has been applied to dormitory allocation. This work shows how assignment mechanisms can be structured around institutional rules such as entrance criteria, priority rankings, capacity limits, and, in later extensions, applications by student groups rather than only by individuals (Perach et al., 2008; Perach and Anily, 2022).

A related perspective comes from the literature on preference aggregation in group recommendation. Although this literature typically examines how items are recommended to groups of joint decision-makers, it is still informative here because it formalizes how an overall evaluation can reflect both aggregate suitability and internal disagreement. Amer-Yahia et al. (2009) model group recommendation through a consensus function that combines relevance and disagreement, while Felfernig et al. (2017) show that the choice of aggregation heuristic depends on the item domain, using shared apartments as an example of a high-involvement setting. These studies therefore do not mirror the present allocation problem directly, but they provide a useful analogy for thinking about how apartment-level scores may need to balance average fit with protection against weak fits.

The aforementioned literature suggests that compatibility-based apartment allocation involves two linked tasks: estimating interpersonal fit and deciding how such information should be aggregated and incorporated into an assignment process. On this basis, the next chapter outlines a narrowed rationale and problem formulation of the study as well as a methodological backbone for attempting to tackle the problem at hand.

3 Rationale and Problem Statement

This section reiterates the central problem addressed in the thesis. It discusses the relevance of tenant compatibility in co-living environments and highlights the limited development of structured approaches for representing and applying compatibility in practice.

3.1 Rationale for the Current Study

Despite the continued growth of co-living, tenant incompatibility remains a recurring operational issue. Allocation decisions are often based on manual judgment, incomplete information, or intervention only after tensions have already emerged. In practice, such approaches rarely capture the multi-dimensional nature of shared living, where routines, habits, preferences, and interpersonal fit can all shape apartment dynamics.

Although compatibility has been examined in related areas such as roommate matching and interpersonal relationships, professionally managed co-living environments differ in important ways. They are often characterized by higher turnover, greater diversity, and stronger operational constraints. These conditions make compatibility more difficult to assess and help explain why structured allocation approaches remain limited in practice.

3.2 Justification for the Study

Commercial co-living places strangers into shared private living environments, yet tenant allocation is still rarely guided by any structured assessment of compatibility. For residents, this creates uncertainty around everyday living quality; for operators, it leaves clear product and differentiation potential underused. This study addresses that gap through a transparent compatibility-based decision-support framework.

3.3 Broader Implications

The study has implications for both research and practice.

Academic relevance: it transfers compatibility modeling into the co-living setting and shows how interpersonal fit can be formalized for analytical use in shared housing.

Practical relevance: it offers operators a structured basis for composing apartments and assigning applicants under vacancy conditions.

More broadly, the framework demonstrates how compatibility can be incorporated into co-living

decisions in a way that is transparent, usable, and closer to the lived reality of shared housing.

3.4 Problem Definition

In commercial co-living, tenant allocation still tends to focus on availability and operational constraints rather than on a structured assessment of interpersonal fit. While prior research has explored compatibility in adjacent contexts, there is limited work on how resident characteristics can be translated into an allocation framework suited to shared apartments.

The central problem addressed in this study is therefore:

How can tenant compatibility in co-living environments be represented through resident-relevant characteristics and translated into a practical framework for apartment allocation?

The study therefore connects compatibility research with an applied allocation setting and focuses on making the concept analytically usable in practice.

3.5 Significance of the Study

The study contributes by treating compatibility in co-living as a structured allocation problem rather than as an informal or intuitive judgment. It proposes a transparent way to organize resident-relevant characteristics into a compatibility framework that can be examined at both the pairwise and apartment level.

From a practical perspective, the study offers a more deliberate basis for composing shared apartments and assigning applicants to vacancies. In doing so, it shows how compatibility can be incorporated into co-living operations in a way that is interpretable, scalable, and relevant to real housing decisions.

4 Research Methodology

This chapter sets out the methodological approach used to develop the compatibility framework. It describes the research design, data sources, preprocessing procedures, and analytical steps applied across the two stages of the study. The aim is to provide a clear and coherent account of how the framework was developed and translated into an operational approach to tenant allocation.

4.1 Research Design

The study is organized in two stages. Stage 1 draws on a large secondary personality dataset to establish a general trait-based foundation. Stage 2 builds on this foundation by incorporating lifestyle and tenancy-related attributes into a compatibility framework tailored to shared housing contexts.

Overall, the design combines the development of a structured tenant representation with compatibility assessment and allocation-oriented application. Drawing directly from literature in the field, the study connects a broader personality-based foundation with context-specific co-living data while maintaining interpretability and methodological consistency.

4.2 Data Sources

The analytical framework draws on two main data sources: a large-scale secondary personality dataset and an operational co-living dataset representing tenants within a shared housing context.

4.2.1 Secondary Data: Personality Dataset

The personality data was obtained from the OpenPsychometrics IPIP-FFM repository and contains more than one million anonymized responses to a Big Five personality inventory. Each observation includes 50 Likert-scale items covering Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness. This dataset serves as the basis for Stage 1 by providing a broad empirical foundation for the construction and analysis of personality-based trait representations.

4.2.2 Co-living Dataset and Synthetic Survey Construction

To capture compatibility-relevant characteristics in a co-living context, a survey instrument was designed to collect tenant-level personality and lifestyle information beyond what was available in the operational housing data (see Appendix A.5). However, participation was too limited to support direct empirical modeling.

To allow the proposed framework to be developed and applied in Stage 2, synthetic responses were constructed for the survey-based variables. These variables were introduced to represent compatibility-relevant aspects of shared living, including lifestyle preferences, behavioral tendencies, and selected personal characteristics, in a form that could be incorporated into the analytical framework. The resulting dataset comprises 563 tenant records across 167 apartments. Accordingly, Stage 2 should be understood as an application-oriented illustration of how such information can be structured and used within a compatibility-based allocation approach under constrained data conditions.

4.3 Data Preparation

4.3.1 Stage 1 – Personality Dataset

Quality Filtering Initial preprocessing was used to retain only valid survey submissions. Records with incomplete responses were removed, and the remaining observations were filtered using the embedded attention check ($IPC = 1$) and completion-time thresholds. More specifically, responses completed in less than 60 seconds and responses above the 99th percentile of completion time were excluded. This was intended to remove implausibly fast submissions as well as unusually long cases.

Feature Construction The 50 IPIP-FFM items were mapped to the five Big Five dimensions using the standard scoring key, with reverse coding applied where required. Item responses were then aggregated into trait-level scores for Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness. These trait measures formed the main feature space for the subsequent analysis.

4.3.2 Stage 2 – Operational and Synthetic Co-living Data

Anonymization and Data Protection The operational tenant dataset was exported from the back-office system of the co-living operator and initially contained personally identifiable and administrative information not required for the analysis. To protect tenant privacy, direct

identifiers and non-essential administrative fields were removed during preprocessing. First names were retained only temporarily for name-based gender inference and were excluded from the final analytical dataset. In addition, apartment and unit codes were normalized and then pseudonymized so that the identifiers used in the analysis do not correspond to the original internal labels.

Cleaning and Harmonization The operational dataset represented tenants living in a Berlin co-living portfolio at the time of extraction in November 2025. After removing non-tenant placeholder entries, the remaining variables were translated and harmonized into a consistent English-language format. Core fields retained for analysis included room unit, apartment unit, move-in date, move-out date, date of birth, nationality, and occupation.

Date variables were parsed into a common format and used to derive age, length of stay, and remaining stay in days relative to the data reference date. Missing age values were imputed using the sample median. Occupation labels were standardized, nationality codes were converted into country labels, and missing categorical values were retained as *Unknown* where appropriate. Because gender was not directly available, it was inferred from first names using a name-based classification procedure. Apartment size was derived from the unit identifier by taking the highest room number observed within each apartment.

After these steps, the operational file was reduced to the variables needed for the compatibility framework, including pseudonymized unit identifiers, demographic attributes, tenancy measures, and apartment size.

Construction and Integration of Survey-Based Variables To represent compatibility-relevant aspects of shared living that were not available in the operational file, a survey instrument was designed to capture personality, lifestyle, behavioral tendencies, and perceived compatibility (Appendix A.5). This survey was transmitted to the exact same tenant population via the internal management system on two separate days in late November 2025.

As participation was extremely low, these survey-based variables were represented through a synthetic dataset, generated through an LLM. This had a major effect on the positioning of the research. Rather than combining literature- and industry-informed modeling with supervised validation, the study was reframed as an exploratory structural compatibility approach with dual operational application visualization tools.

The synthetic file structure is consistent with the survey instrument and included an abbreviated personality battery with two IPIP items per Big Five dimension, together with lifestyle and household-related variables such as sleep schedule, noise sensitivity, cleanliness, guests, cooking behavior, smoking, alcohol use, social preference, chores, conflict style, and perceived

compatibility within the apartment. The decision to include only two IPIP items per Big Five dimension, rather than ten as in the Stage 1 dataset, was made to reduce respondent fatigue and lower the barrier to participation, although this ultimately did not translate into meaningful response uptake. The abbreviated IPIP items were aggregated into Big Five trait scores in a manner consistent with Stage 1. In addition, ordinal and Likert-type responses were converted into numeric form where required so that they could be incorporated into the modeling framework.

The operational tenant file and the synthetic survey-based file were then combined into a single analytical dataset. This merge was carried out using apartment identifiers together with a within-apartment row index so that each tenant record could be linked to one set of survey-based attributes within the same apartment structure. The resulting dataset served as the basis for the Stage 2 analysis.

4.4 Conceptual Compatibility Framework

Compatibility is conceptualized as relational fit between residents based on selected observable characteristics. Rather than treating compatibility as a single directly observed attribute, the study models it as a structured pairwise construct derived from resident profiles and then translates it into apartment-level evaluation.

In Stage 1, each resident is represented through Big Five personality traits. Pairwise compatibility is computed from trait-level rules that are primarily similarity-based, while allowing selected traits to follow a complementarity logic where theoretically justified. Trait-specific weights are used to reflect different co-living priorities across model specifications. The resulting pairwise compatibility matrix then serves as the basis for simulated apartment formation and for comparison with random baseline assignments across typical co-living group sizes. Because these simulations require repeated access to the same compatibility structure, a cached matrix backend is used to improve computational efficiency.

Stage 2 retains the pairwise logic but does not rely on a single extended feature vector. Instead, it combines two separate compatibility layers. The first is a personality layer, built from the same Stage 1 personality engine and selected for each applicant according to the stated flatmate priority profile. The second is a lifestyle layer based on numerically encoded lifestyle variables. These lifestyle inputs are not globally standardized beforehand; instead, absolute pairwise differences are converted into similarity scores within the compatibility function using feature-specific theoretical scale ranges. The two layers are then combined into an applicant-conditioned final pairwise matrix using applicant-specific weights.

The Stage 2 application is vacancy-based. Vacant units are derived from the observed apartment structure by identifying missing unit positions within apartments. For each candidate vacancy,

the focal applicant is evaluated together with the current residents of that apartment. Pairwise compatibility is therefore translated into apartment-level structural measures such as overall harmony, fairness, and minimum pair compatibility. These scores are further supplemented by contextual apartment-level flags, allowing the final recommendations to remain structured, interpretable, and operationally usable.

4.5 Computational Design and Subsampling Strategy

Compatibility modeling relies on pairwise comparisons between individuals, which increase quadratically with sample size. To keep Stage 1 computationally tractable while preserving the overall structure of the personality space, model development was conducted on a reproducible random subset of 2,000 observations. Descriptive exploration and initial structural assessment, however, were based on the full cleaned dataset.

This design reflects the progression of the study from dyadic compatibility estimation to group-level application. Pairwise scores were first developed and examined as the core analytical unit, and were subsequently used within grouping and assignment procedures.

5 Model Development

5.1 Stage 1 – Personality-Based Compatibility

5.1.1 Trait Representation

In Stage 1, each individual is represented through the Big Five trait scores derived from the IPIP-FFM inventory. These scores constitute the core input for the personality-based compatibility framework.

Each trait score is computed as the mean of its corresponding keyed questionnaire items after reverse-coding the negatively formulated items. The resulting values remain on the original 1–5 Likert scale, which preserves direct interpretability and ensures that pairwise differences can be evaluated on the underlying trait metric.

The five traits considered are Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness. Together, they provide a compact personality representation that forms the basis for all pairwise compatibility calculations in Stage 1.

This trait-based representation enables individuals to be compared systematically and serves as the foundation for the compatibility functions introduced in the following sections.

5.1.2 Exploratory Distance/Similarity Metrics and Structural Inspection

Before specifying the compatibility score, the personality space was explored using several geometric proximity measures. Pairwise Euclidean, Manhattan, and cosine distances were computed on the raw Big Five trait scores, since all five traits were measured on the same bounded scale. Cosine distance on the raw representation was heavily compressed and therefore of limited interpretive value. Cosine distance was also inspected after standardization.

To evaluate whether alternative distance concepts identified broadly similar high-proximity pairs, Jaccard similarity was used to compare the overlap among the top 25 pairs generated by the Euclidean and Manhattan distance measures.

These exploratory metrics served only a diagnostic purpose. They were used to characterize the geometry of the personality space and to assess whether different proximity concepts yielded broadly consistent patterns. In addition, clustering through silhouette analysis and dimensionality reduction through principal component analysis were briefly explored to better understand the broader structural features of the data.

5.1.3 Compatibility Function Formulation

The Stage 1 compatibility score was designed as a transparent, trait-separable function based on absolute pairwise differences in Big Five trait values. A Manhattan-style logic was preferred over a single aggregate geometric distance because it preserves the contribution of each trait individually and makes later weighting and sensitivity adjustments straightforward. Since all trait scores were measured on the same bounded 1–5 scale, absolute differences could be interpreted directly without further rescaling at this stage.

For any two individuals i and j, each trait first contributes a bounded compatibility component on a 0–1 scale. Under a similarity logic, compatibility decreases linearly as the absolute trait difference increases:

$$c_t^{\text{sim}}(i, j) = 1 - \frac{|x_{it} - x_{jt}|}{\Delta_{\max}},$$

where $\Delta_{\max} = 4$ denotes the maximum possible difference on the original scale. This implies a score of 1 for identical trait values and 0 for the largest possible difference.

To allow for the possibility that moderate differences may be beneficial on selected traits, the same framework can also apply a complementarity component. In that case, the highest compatibility is obtained not at zero difference, but at a predefined ideal distance δ^*:

$$c_t^{\text{comp}}(i, j) = 1 - \frac{\big||x_{it} - x_{jt}| - \delta^*\big|}{\max(\delta^*, \Delta_{\max} - \delta^*)}.$$

This formulation preserves the 0–1 scale while allowing compatibility to peak at a moderate difference rather than exact similarity.

The overall pairwise compatibility score is then computed as a weighted average across the included traits:

$$C_{ij} = \frac{\sum_{t \in S} w_t \, c_t^{\text{sim}}(i, j) + \sum_{t \in K} w_t \, c_t^{\text{comp}}(i, j)}{\sum_{t \in S} w_t + \sum_{t \in K} w_t},$$

where S denotes the set of similarity-based traits, K the set of complementarity-based traits, and w_t the trait-specific weights. Because the function is normalized by the total weight sum, the weights affect the relative influence of traits rather than the scale of the final score itself.

In the main Stage 1 reference specification, all five personality traits were treated as similarity-based and assigned equal weight. This produces the most conservative and interpretable version of the model, in which compatibility is highest when two individuals are broadly alike across the Big Five profile. Alternative weight structures and mixed similarity–complementarity specifications were implemented only as sensitivity analysis in order to examine how strongly the resulting compatibility patterns depend on these assumptions.

5.1.4 Pairwise Compatibility Matrix Construction

For each specification, the compatibility function was evaluated for all pairs of individuals in the Stage 1 modeling subset, yielding a symmetric pairwise compatibility matrix. Each off-diagonal entry represents the compatibility score of a given pair, while diagonal entries were set to one by definition.

This matrix formed the basis for subsequent comparison across specifications, including differences in score distributions and the extent to which alternative parameter choices preserved the relative ranking of candidate matches.

5.1.5 From Pairwise Compatibility to Group Assignment

Pairwise compatibility scores were used as the basic input for constructing and evaluating full group assignments. For any candidate group, the code first derives a member-level utility for each person, defined as that member's average compatibility with all other members of the same group. Group quality is then summarized from three complementary perspectives. Harmony captures the average of these member-level utilities and therefore reflects the overall level of fit within the group. Fairness is defined as the lowest member utility in the group and is intended to highlight whether one person is substantially less well matched than the others. In addition, the minimum pairwise compatibility within a group was tracked as a diagnostic safeguard against constellations that perform reasonably well on average but still contain a particularly weak dyadic match.

To move from pairwise compatibility to full assignments, the Stage 1 notebook implements a compatibility-guided greedy heuristic. The modeling subset is partitioned into feasible group sizes of three or four, after which groups are built sequentially from the remaining pool. For each new group, the algorithm first selects a seed from among the strongest remaining candidates, where strength is approximated by average compatibility to the remaining pool. It then adds further members one at a time by evaluating, for every available candidate, how the inclusion of that person would update the current group's quality. The candidate with the best local objective value is chosen at each step. This objective combines harmony and fairness, allowing the procedure to prefer assignments that are not only strong on average but also avoid leaving one member with a clearly weaker fit.

Because greedy solutions depend partly on initialization and group-size order, the procedure is repeated across multiple random restarts and the best full assignment is retained. In the reference experimental setup, this heuristic was benchmarked against repeated random assignments generated under the same group-size constraints. Assignment quality was then compared at several levels: across the full assignment, across groups, and across individual members. The

reported summaries include overall harmony, overall fairness, lower-tail utility and pairwise statistics, and the minimum observed pair compatibility, thereby allowing comparison not only of average performance but also of how evenly compatibility is distributed across the resulting allocation.

5.2 Stage 2 – Extended Compatibility Framework

5.2.1 Feature Groups and Model Inputs

Stage 2 combines four feature groups. First, aggregated Big Five trait scores from the survey-based extension provide the personality input, scored using the same compatibility structure introduced earlier but on a different dataset. Second, encoded lifestyle variables capture everyday co-living preferences, including sleep schedule, noise sensitivity, social vibe, cleanliness, guests, alcohol use, smoking, cooking at home, and chores. Third, applicant-side preference variables — flatmate priority and compatibility importance — personalize the weighting of the model layers. Fourth, selected apartment and resident context variables from the operational data support vacancy-level structural scoring and contextual flagging, including apartment size, age, gender, nationality, occupation, and remaining stay duration.

5.2.2 Personality Compatibility Layer

The personality layer is implemented through a reusable scoring module that carries the Stage 1 logic into the Stage 2 recommendation framework. For the Stage 2 population, three pairwise personality compatibility matrices are precomputed: *equal*, *harmony*, and *reliability*. All three profiles use an all-similarity specification across the Big Five dimensions, so the only difference between them lies in the relative trait weights.

The equal profile assigns weight 1.0 to each trait. The harmony profile places greater emphasis on Agreeableness and Emotional Stability (1.5 each), followed by Conscientiousness (1.2), while giving slightly less weight to Extraversion (0.8) and Openness (0.7). The reliability profile places the strongest emphasis on Conscientiousness (1.7), with secondary emphasis on Agreeableness and Emotional Stability (1.2 each), and lower weight on Extraversion and Openness (0.7 each). These matrices are computed once in advance and then selected at recommendation time.

The applicant's stated flatmate priority determines which profile is applied. Priorities related to reliability in shared spaces, respect for boundaries, or similar daily routines activate the reliability matrix. Priorities related to being social and engaging or shared interests and personality fit activate the harmony matrix. Flexible or non-committal responses default to the equal matrix. This introduces applicant-level personalization while keeping the underlying personality scoring

logic fixed.

5.2.3 Lifestyle Compatibility Layer

The lifestyle compatibility layer is implemented as a separate pairwise matrix based on nine encoded co-living preference variables: sleep schedule, noise sensitivity, social vibe, cleanliness, guests, alcohol use, smoking, cooking at home, and chores. For each feature, the absolute difference between two tenants is transformed into a bounded similarity score by dividing by that feature's observed span in the Stage 2 sample and reversing the scale so that higher values indicate greater similarity.

The pairwise lifestyle score is then computed as the equally weighted average across all available lifestyle features. Repeating this procedure for all tenant pairs yields a symmetric lifestyle compatibility matrix. Some additional encoded survey variables were retained elsewhere in the dataset, but excluded from this core layer.

5.2.4 Applicant-Personalized Combined Compatibility Layer

The applicant-personalized compatibility layer combines the selected personality matrix and the lifestyle matrix into a single final pairwise representation. This is implemented as a convex combination of both layers. The choice of personality profile is governed by the applicant's flatmate priority, while the layer weights are determined by the reported importance of interpersonal compatibility.

Rather than fixing these weights globally, the model rescales the applicant's compatibility-importance response over its observed range in the Stage 2 sample and maps it to a personality weight between 0.40 and 0.70. The lifestyle weight is defined as the remainder to one. As a result, applicants who place greater emphasis on compatibility receive a more personality-driven final score, whereas lower importance shifts the balance toward lifestyle fit. If no valid importance value is available, the model defaults to a 0.55/0.45 split between personality and lifestyle.

The resulting combined matrix remains on the same bounded scale as the underlying layers and provides the central pairwise input for the structural vacancy scoring step.

5.2.5 Vacancy-Level Structural Scoring

For each vacancy, the applicant is inserted into the corresponding apartment and evaluated jointly with the current residents using the final combined compatibility matrix. This produces an

apartment-specific compatibility submatrix from which member utilities and structural summary metrics are calculated.

Harmony is defined as the mean of all member utilities, where each member utility corresponds to that person's average compatibility with the other apartment members. Fairness is the minimum member utility and captures the weakest individual outcome within the proposed composition. The minimum pairwise compatibility is additionally recorded to reflect the weakest dyadic link in the apartment.

The final structural vacancy score is computed as a weighted average of these three metrics, with weights of 0.45 for fairness, 0.35 for harmony, and 0.20 for minimum pairwise compatibility. Vacancies are ranked by this score, with fairness and harmony used as secondary ordering criteria.

5.2.6 Context Flags and Interpretive Information

Each vacancy is accompanied by a small set of contextual flags that are attached after scoring and shown as interpretive notes rather than incorporated into the structural score. These flags are intended to make the recommendation output easier to read without changing the compatibility ranking itself.

The planned flags describe selected features of the current resident composition relative to the applicant. They indicate whether the nearest age match in the apartment is still more than five years older or younger, whether a high-turnover situation is present, and whether the resident group is compositionally homogeneous by gender, nationality, or occupation. For gender and occupation, the note can explicitly state the dominant resident composition, whereas nationality homogeneity is indicated only in general form. These indicators are therefore best understood as contextual reading aids rather than scoring components.

5.2.7 Recommendation and Matching Logic

The Stage 2 recommendation procedure is implemented as an applicant-conditioned vacancy ranking. For a selected applicant, the model first determines the appropriate personality profile from the stated flatmate priority and then combines this personality matrix with the lifestyle matrix using applicant-specific layer weights derived from the reported importance of compatibility. This yields a final pairwise compatibility matrix tailored to the applicant.

Each vacancy is then evaluated by inserting the applicant into the corresponding apartment and computing the structural vacancy score for the resulting resident composition. Vacancies are ranked in descending order of this score, with fairness and harmony used as secondary ordering

criteria. Additional controls can further restrict the output, including stricter minimum-fit thresholds and the optional exclusion of vacancies with selected contextual flags.

The model outputs are implemented in two complementary interfaces with different analytical purposes. The primary application is an applicant-facing vacancy recommender, in which the applicant-conditioned structural score is translated into ranked apartment suggestions. For each vacancy, the interface displays the aggregated match result derived from the underlying compatibility layers, structural apartment-level scoring, and contextual apartment flags. In parallel, a separate operator-facing dashboard applies the same compatibility logic in a fixed, non-personalized form to the current resident composition of occupied apartments. Rather than generating applicant-specific recommendations, this interface supports apartment-level monitoring by exposing composition scores, resident-level utilities, pairwise compatibility relations, and apartment-level comparison fields across the portfolio.

6 Results and Evaluation

This chapter examines how the compatibility framework performs across both modeling stages. In the absence of real observed compatibility outcomes, the evaluation focuses on the patterns in the resulting scores, their internal consistency, and their usefulness for allocation decisions.

6.1 Stage 1 – Personality-Based Compatibility

6.1.1 Exploration of Personality and Distance Structure

The Big Five traits showed only weak to moderate associations with one another, with no correlation above 0.31 and the highest value observed between extraversion and agreeableness. The pairwise plots also did not reveal strong linear patterns or clearly separated clusters. The PCA and silhouette diagnostics support the same reading, suggesting a relatively continuous personality space rather than a sharply segmented one (Appendix, Figures 5, 7, and 8).

This structure remained fairly stable across the main distance choices considered in Stage 1. Euclidean and Manhattan distance were highly aligned, with Pearson and Spearman correlations of about 0.97 and a mean Jaccard overlap of 0.646 for top-25 neighbour sets. The cosine-based variant was somewhat less aligned, but still showed substantial agreement with the other two specifications, with correlations between 0.73 and 0.80. This suggests that changing the distance formulation had limited effect on which profiles were identified as the closest matches.

6.1.2 Compatibility Score Distribution and Sensitivity

Pairwise compatibility scores remained on a very similar overall scale across the main Stage 1 weighting specifications, with mean values between 0.799 and 0.802. The underlying score structures were also highly correlated, with pairwise correlations between 0.9566 and 0.9737. This suggests that moderate changes in trait weighting altered scores only slightly in aggregate.

The more important question was whether these alternative specifications changed the ranking of potential matches. Relative to the reference model, mean Kendall's tau values ranged from 0.322 to 0.844 across the broader model suite. Rank agreement remained high when only weights were adjusted or when complementarity assumptions were kept mild, but declined more clearly as stronger ideal-delta assumptions were introduced. In practical terms, this means that the main Stage 1 matching patterns were fairly robust to modest specification changes, while more assertive complementarity settings had a more noticeable effect on who was identified as the best match (Appendix, Figures 16, 17, and 18).

6.1.3 Group Formation and Assignment Simulation

The compatibility-guided assignment scored better than the random baseline across all reported criteria. At the assignment level, harmony increased from 0.780 to 0.937, fairness from 0.488 to 0.698, and minimum pair compatibility from 0.378 to 0.650. Lower-tail outcomes improved as well, with the 10th percentile of resident utility rising from 0.706 to 0.903 and the 10th percentile of pair compatibility from 0.671 to 0.898. The corresponding boxplots show that this was not only an average effect, but a broader upward shift in group-level and resident-level outcomes, especially for fairness, weakest-pair protection, and member utility (Appendix, Figures 19, 20, and 21).

At the same time, this result should be read as a quick internal check rather than as evidence of directly achievable real-world allocations. Since the same compatibility logic is used to build and evaluate the groups, an improvement over random assignment is expected. In addition, the greedy full-assignment setup abstracts from practical operational constraints, so these exact formations would not arise in practice. Its value here is mainly to show that the scoring framework behaves as intended and can construct systematically stronger groups under its own objective.

6.2 Stage 2 – Extended Compatibility Framework

6.2.1 Extended Feature Space Overview

The Stage 2 feature space extends the Stage 1 personality layer with operational resident and apartment context. On the observed-data side, the sample is concentrated in larger shared units, with 4-room apartments clearly dominating at about 250 observations, followed by 3-room units at roughly 100 and 5-room units at just under 90. Very small or very large units are much less common, so the recommendation setting is primarily shaped by mid-sized co-living apartments rather than by edge cases (Appendix, Figure 22).

The resident population is also concentrated in a fairly specific demographic segment. Ages cluster mainly in the early to mid-20s, with the highest density around 23–24 and a visible but thinning right tail into the 30s and 40s (Appendix, Figure 24). Occupation is dominated by students at around 330 observations, followed by employed residents at roughly 145, while all other categories are comparatively small (Appendix, Figure 27). Gender is led by male entries at about 260, compared with roughly 170 unknown and 130 female cases, and the nationality distribution is internationally mixed, with India, Germany, unknown, and France forming the largest groups at around 90, 55, 53, and 43 observations respectively (Appendix, Figures 25 and 26). From an evaluation perspective, this suggests that the Stage 2 recommendation setting is

grounded in a young, student-heavy, internationally mixed co-living population, although the sizeable "unknown" categories mean that demographic context flags remain somewhat coarse.

The remaining lifestyle and preference variables, including sleep schedule, noise sensitivity, cleanliness, guests, alcohol use, smoking, cooking at home, chores, flatmate priority, compatibility importance, and synthetic compatibility ratings, are reported in the Appendix (Figures 29, 30, 31, and 28). These distributions should not be interpreted too strongly, since they are based on synthetic survey-style inputs and are included mainly to illustrate how such variables could be integrated into the broader Stage 2 representation.

6.2.2 Stage 2 Recommendation Setting

The Stage 2 recommendation exercise was carried out on a real vacancy set of 47 vacant units across 40 apartments. Relative to the broader Stage 2 stock of 167 apartments, this is a fairly narrow operational choice set rather than a fully flexible matching problem. Most vacant apartments contributed a single vacancy, while a smaller number contributed two, which also explains why some room-level recommendations repeat the same apartment-level compatibility values.

This is useful context for interpreting the results. The model is not constructing entirely new flat shares from scratch, but ranking a limited set of existing vacancies against the resident compositions already in place. In that sense, the Stage 2 output is closer to a real booking-support scenario than the Stage 1 assignment simulation.

6.2.3 Applicant-Facing Recommendation Output

For the illustrated applicant, the highest-ranked vacancies were concentrated in a relatively narrow upper band, with the best structural vacancy score at 0.816 and the top ten mostly falling between about 0.779 and 0.816. This suggests that the recommendation problem is not driven by one overwhelmingly dominant option, but by several reasonably strong candidates that still differ in the details of their composition.

Those differences are exactly where the apartment-level metrics become useful. Vacancies with similar overall averages can still separate once fairness and minimum-pair compatibility are considered. In the prototype view, this becomes interpretable at the applicant level. The top visible recommendation is shown at 89/100 and the next visible option at 83/100, but the added value is not the score alone. The interface translates the ranking into concrete reasons for and against a booking decision, such as full agreement on noise sensitivity, smoking, and chores in the strongest example, alongside caution flags on age difference and short remaining stays within the flat.

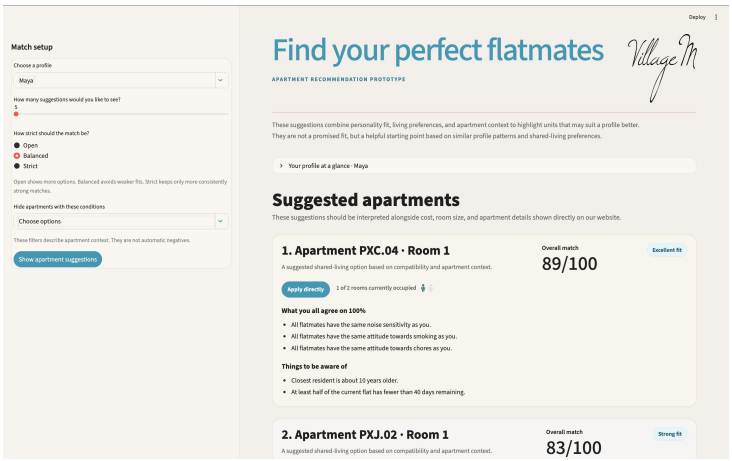

Figure 1: Applicant-facing recommendation prototype

From a decision-support perspective, this makes the output more useful than a raw ranking table. It helps applicants identify promising options, but also understand where a seemingly strong match may still come with trade-offs that matter for the final booking decision.

6.2.4 Operator Dashboard and Portfolio Monitoring

The operator-facing view shifts the focus from personalized vacancy ranking to monitoring the current quality of existing flat compositions. In the illustrated dashboard, 167 apartments are tracked, of which 1 is currently flagged as a potential issue and 6 are marked as very strong, with an average composition score of 74/100. This already shows that the distribution is not flat: most apartments fall into a middle range, while only a small number stand out clearly at either end.

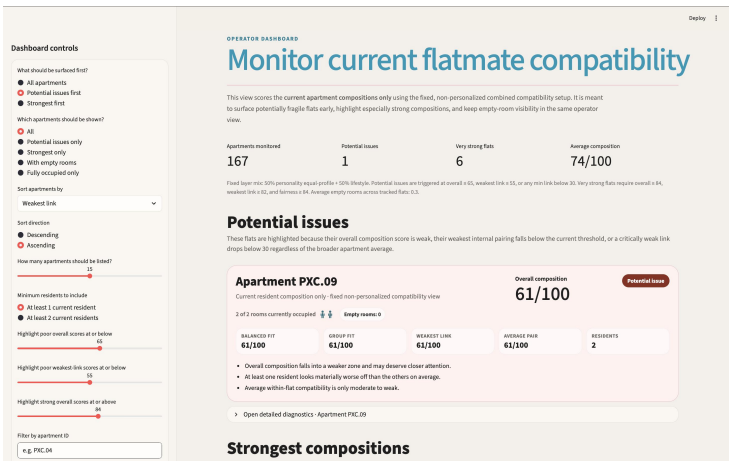

Figure 2: Operator-facing apartment compatibility dashboard.

The practical value of this view lies in triage and oversight. Instead of reviewing apartments one by one, operators can surface weaker constellations quickly, sort by weakest link or overall

composition, and keep empty-room visibility in the same interface. The example shown for Apartment PXC.09 illustrates this well: the flat is flagged at 61/100, with weak values across the key composition indicators, making it a reasonable candidate for closer review. This should not be interpreted as a prediction of conflict, but it does provide an early warning signal that the current composition may be more fragile than average.

Operationally, this creates two kinds of value. Strong flats become easier to recognize and potentially easier to market with confidence, while weaker flats can be reviewed earlier before another placement is made into an already fragile composition. In that sense, the dashboard is less about automation and more about improving visibility across the portfolio.

7 Challenges and Limitations

7.1 Data Limitations

The study was constrained by substantial limitations in the available empirical data. The most important issue was the absence of a suitable ground truth for compatibility. No systematic historical data were available on successful or unsuccessful tenant matches, and no validated compatibility outcomes existed that could be used to directly assess whether particular tenant constellations led to better shared living outcomes. This meant that compatibility could not be observed in a way that would allow straightforward empirical validation from existing datasets in this specific domain.

A second limitation concerns the level at which compatibility was intended to be captured. The original survey design aimed to measure compatibility between specific co-residents, which would have aligned more closely with the interpersonal nature of the concept. In practice, this was not feasible, as asking tenants to evaluate identifiable flatmates raised practical and ethical concerns by the co-living operator. The survey was therefore limited to apartment-level assessments of overall compatibility. As a result, the data would have reflected aggregated perceptions of harmony within a shared living environment rather than direct pairwise compatibility between individuals.

The available survey data also relied entirely on self-reported information. This applied both to respondents' descriptions of their own living habits and preferences and to their assessments of compatibility in the apartment. Such responses are inevitably imperfect. They may be shaped by social desirability, selective recall, or temporary experiences within the household. The temporal variables were intended to control for the latter. Behaviors that are especially relevant in co-living, such as cleanliness expectations, sensitivity to noise, or social boundaries, may therefore not be reported fully or truthfully.

In addition, the number of complete survey responses remained limited. This restricted the analytical depth of the dataset and left too little empirical basis for deriving a robust outcome variable from observed compatibility assessments, consistent with prior research in this field. To extend the feature space in Stage 2 and test the broader framework, synthetic survey data were introduced. This made it possible to continue the structural modeling exercise and develop the prototype further, but it also reduced the extent to which resulting patterns could be read as evidence of actual compatibility dynamics. The Stage 2 framework should therefore be understood primarily as a methodological proof of concept rather than as a validated predictive system. Hence, attention throughout this thesis shifted towards that and only alluded to the survey outcome variables in very limited and careful fashion.

Beyond the dataset used in the thesis, the practical implementation of such a matchmaking

approach would raise further data collection challenges. A system of this kind would depend on both applicants and existing tenants providing personality- and lifestyle-related information in a sufficiently complete and consistent manner. In practice, such participation may be uneven, selective, or strategically biased. For the framework to function reliably, participation would likely need to be widespread across both groups, which may be difficult to achieve in an operational setting.

7.2 Methodological Constraints

These data limitations shaped the methodological design of the study in important ways. Most notably, the project could not be framed as a conventional supervised prediction task. A stronger predictive design would have required observed compatibility outcomes at sufficient scale and at the relevant level of analysis. Since neither condition was met, the study could not train and test models against a reliable labeled target and apply various machine learning models to accurately predict compatibility.

Instead, the framework was developed as a theory-informed and structurally derived compatibility model. Existing literature on interpersonal fit, personality, and shared living provided the basis for defining relevant dimensions of compatibility, while the empirical data served mainly to operationalise and extend this logic. The resulting approach is therefore better understood as an analytical decision-support framework and step towards improving the tenant experience and community operator's community management potential in an innovative way.

Though to reiterate, no formal validation of predictive accuracy was possible. The generated compatibility scores should not be interpreted as empirically verified predictions of co-living success, but as structured estimates designed to support allocation decisions under conditions of limited data.

Finally, the framework was developed within one specific co-living context and around data structures linked to a particular operator. Its assumptions, variables, and outputs may therefore not transfer directly to other housing settings without adaptation and renewed empirical assessment.

7.3 Summary

Overall, the study is constrained by the absence of suitable real-world compatibility outcomes, the reliance on self-reported and apartment-level survey data, and the use of synthetic augmentation in Stage 2. These limitations prevented the development of a fully supervised and empirically validated predictive model. The study should therefore be understood primarily as a literature-

informed compatibility framework and exploratory decision-support prototype for co-living allocation, rather than as a validated predictive allocation system.

8 Contributions

8.1 Theoretical Contributions

This study contributes to the emerging discussion on compatibility in shared living by extending interpersonal compatibility research to the context of commercial co-living.

Its main theoretical contribution is a shift in perspective from dyadic matching to apartment-level compatibility in managed multi-tenant settings. Compatibility is therefore treated not only as a relation between two individuals, but as a question of how a prospective resident fits into an existing household composition.

The study also adopts a broader view of compatibility by combining personality-related fit with lifestyle preferences and selected tenancy-related and sensitive demographic context. This moves the concept beyond narrow demography-omly matching toward a more multidimensional understanding of shared living suitability.

Collectively, these contributions frame compatibility in co-living as a group-level and context-sensitive concept rather than a purely dyadic one.

8.2 Methodological Contributions

Methodologically, the study develops a staged compatibility framework for apartment allocation in co-living environments.

The framework distinguishes between a foundational personality compatibility layer and a second-stage extension that incorporates lifestyle and household preference variables. This preserves a stable interpersonal core while allowing the recommendation logic to reflect practical aspects of shared living.

Compatibility is operationalized at the apartment level rather than through isolated pairwise comparisons alone. Applicants are evaluated against the current resident group of each vacant apartment, which aligns the framework more closely with the allocation setting it is intended to support.

The framework further combines transparent scoring components, including weighted personality compatibility, lifestyle similarity, apartment-level structural evaluation, and contextual flags. The result is an interpretable and adjustable decision-support logic for apartment allocation.

8.3 Practical Contributions

The study also offers a practical contribution for co-living operators.

Most directly, it provides a working prototype that translates compatibility information into ranked apartment suggestions for applicants and a compatibility monitoring dashboard for operators. This creates a more structured basis for allocation decisions and community management, rather than reliance on intuition or informal judgment alone.

In addition, the framework is transparent and configurable. Its main components can be inspected and adjusted, which makes the recommendation logic easier to interpret in an operational setting.

The study also shows how compatibility outputs can be communicated through ranked recommendations and simple interpretive indicators, linking the analytical model to day-to-day housing decisions without displacing human judgment.

8.4 Educational and Curricular Relevance

From an educational perspective, the study reflects several core elements of the MSc in Business Analytics. It required substantial data preparation, feature engineering, and programming in Python, and involved the integration of behavioral, survey-based, and operational housing data into a structured analytical framework.

In that sense, the thesis is most directly linked to the business analytics objective of using transparent and interpretable data to support allocation decisions while also giving applicants structured insight into the living environments they are considering. More cautious links can be made to areas such as predictive analytics, experimentation, data visualization, system integration, and data privacy. These themes are not developed fully within the scope of the study, but they form part of the broader analytical context in which the framework sits.

9 Recommendations and Future Work

9.1 Recommendations for Co-Living Operators

The results suggest that compatibility-based matchmaking may be especially valuable as an applicant-facing decision-support tool. Presented in this way, the framework could offer prospective tenants a more engaging and informative apartment search experience by highlighting not only the monthly rental fees, size, location, amenities and availability, but also aspects of household fit and living style. From an operator perspective, this could create a more distinctive and innovative allocation process while potentially supporting better-informed placement decisions.

From the operator's perspective, a dashboard highlighting potentially problematic or strong compositions could allow for better community management practices and overall oversight. As an example, when conducting viewing appointments, operators could more easily identify which apartments may provide a more positive viewing experience due to harmonious apartment compositions.

For both applications however, outputs should be communicated cautiously. Compatibility scores should not be presented as definitive predictions of future household success, but rather as structured indicators intended to support reflection and comparison. Given the limited empirical validation of the framework, any applicant-facing use would need to remain transparent about this exploratory character.

The recommend-er framework may also hold value as an internal operator tool. Used in this way, it could provide an additional reference point when assessing how well a prospective resident may fit into an existing apartment, complementing rather than replacing professional judgment. Compatibility indicators could further support ongoing housing management by helping operators monitor apartment compositions and identify households in which tensions or mismatches may be more likely to arise.

Any use of demographic information should remain cautious, transparent, and explicitly justified. Variables such as age range or gender composition may in some cases be operationally relevant, but they should only be incorporated through clearly defined rules and should never become implicit optimization targets. Likewise, contextual flags related to sensitive attributes should only be displayed in aggregated rather than individually identifiable form, due to privacy concerns and their regulations.

9.2 Model Enhancements

The most important next step would be the development of a stronger empirical foundation. With richer real-world data on tenant characteristics, apartment compositions, and compatibility outcomes, the framework could move beyond a structurally derived scoring logic toward a more statistically grounded predictive approach.

In particular, repeated compatibility ratings over time would make it possible to construct labeled outcome variables and apply supervised machine learning methods. Instead of relying primarily on literature and industry-informed structuring, future work could train models to estimate compatibility outcomes directly from personality, lifestyle, and tenancy-related features. This would allow alternative modeling approaches to be compared systematically and evaluated against observed outcomes using standard predictive performance criteria.

A richer dataset would also support more robust statistical analysis of which variables matter most, how stable compatibility assessments remain over time, and whether specific combinations of resident characteristics are associated with better housing outcomes. In this way, future versions of the framework could be calibrated and validated on a substantially stronger empirical basis.

9.3 Future Research Directions

Several broader research questions follow from this study. One concerns the extent to which specific resident characteristics actually matter for successful shared living outcomes. Investigating this more directly would strengthen the conceptual basis of compatibility modeling beyond the present framework.

A second direction relates to longer-term housing outcomes such as resident satisfaction, perceived household harmony, complaints, or retention. Longitudinal research would be especially valuable here, as it could show whether early compatibility assessments remain stable over time and whether they are associated with later outcomes.

A further issue is the balance between cohesion and diversity in shared living environments. While a certain degree of similarity may support predictability and ease in everyday interaction, diversity may also contribute to resilience, learning, and community richness. Understanding how this balance should be handled remains an important challenge for compatibility-based allocation approaches.

9.4 Concluding Remarks

This study has shown how tenant compatibility in co-living can be translated into a structured and transparent allocation framework. Although the empirical basis for validation remains limited, it demonstrates the feasibility of combining personality-related information, lifestyle preferences, and apartment-level context in an interpretable and operationally relevant way.

Its main contribution lies in formalizing compatibility as an apartment-level and context-sensitive concept rather than reducing it to demographic matching or isolated pairwise judgments. In this way, the study provides a practical foundation for more structured tenant placement in shared living environments.

At the same time, the framework is not intended to replace human decision-making. Compatibility analytics can add structure and consistency to allocation processes, but final decisions still depend on contextual judgement, operational experience, and responsible oversight.

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Appendix

This appendix includes supplementary visualizations that support the analysis presented in the main results chapter. Figures are ordered to follow the overall structure of the empirical analysis, beginning with Stage 1 descriptive exploration, followed by metric comparison and robustness checks, and concluding with additional Stage 2 descriptive results.

The survey pdf can be found at the end of the appendix.

A.1 Stage 1: Descriptive Exploration

The following figures provide additional descriptive views of the Big Five trait space used in Stage 1, including trait distributions, correlations, pairwise relationships, and lower-dimensional representations.

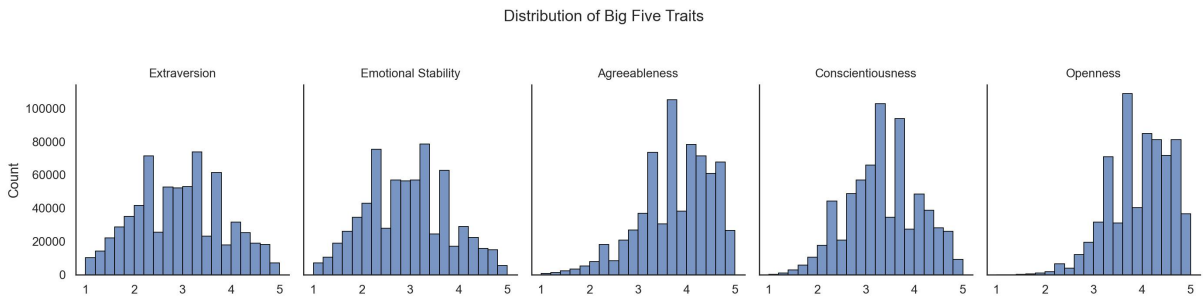


Figure 3: Distribution of Big Five trait scores in the Stage 1 sample.

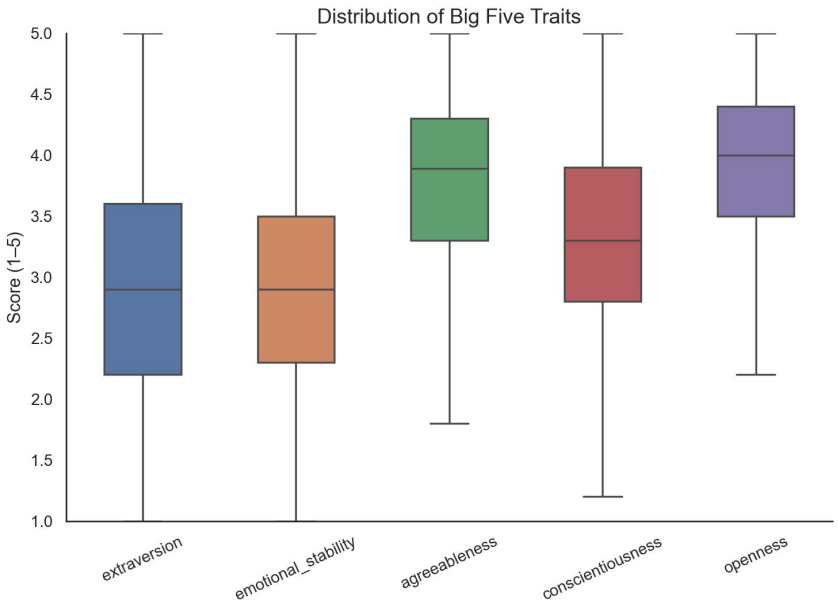


Figure 4: Boxplot summary of Big Five trait distributions in the Stage 1 sample.

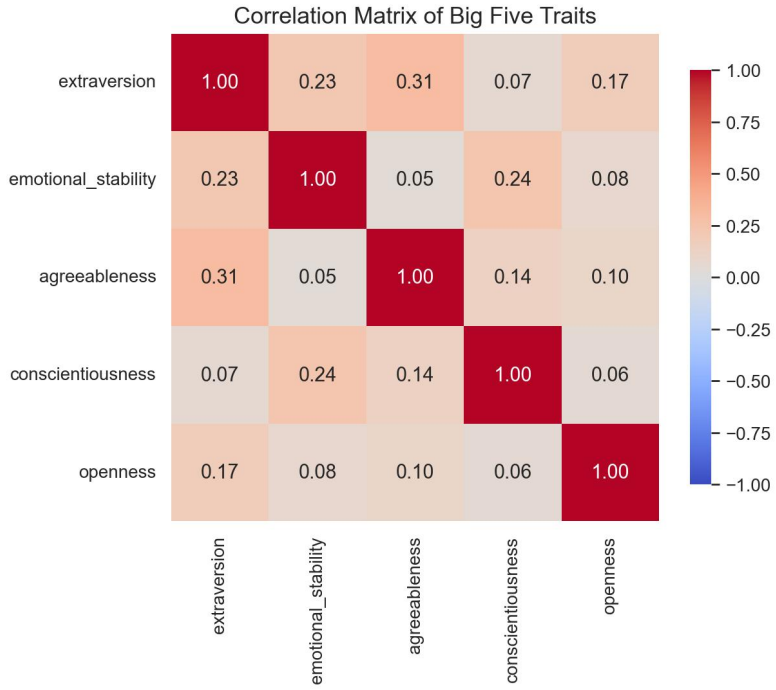


Figure 5: Correlation matrix of the Big Five trait dimensions used in Stage 1.

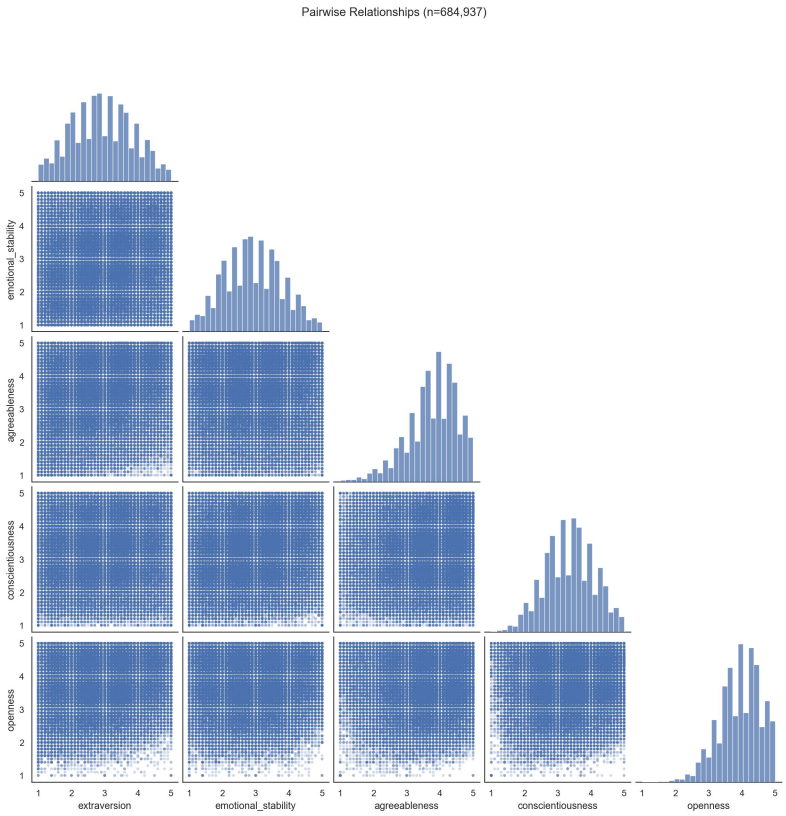


Figure 6: Pairwise relationships between the Big Five trait dimensions in Stage 1.

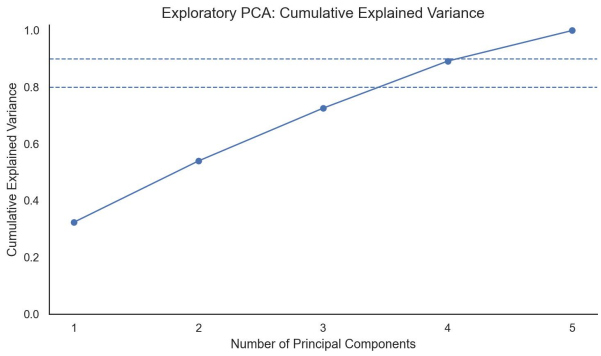


Figure 7: Principal component representation of the Stage 1 personality space.

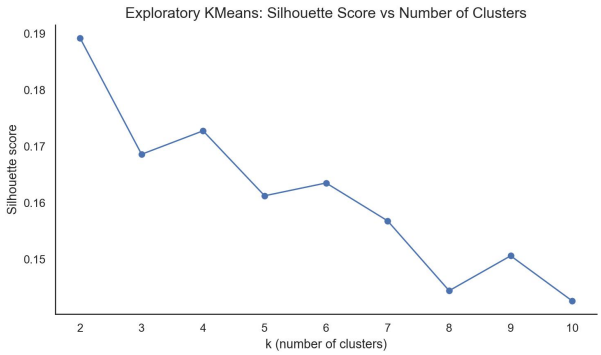


Figure 8: Silhouette analysis used to assess clustering structure in Stage 1.

A.2 Stage 1: Distance Metrics and Compatibility Structure Scores

The next set of figures reports supplementary results on the behavior of alternative distance and similarity measures explored before model development and on the compatibility scores across weighting specifications.

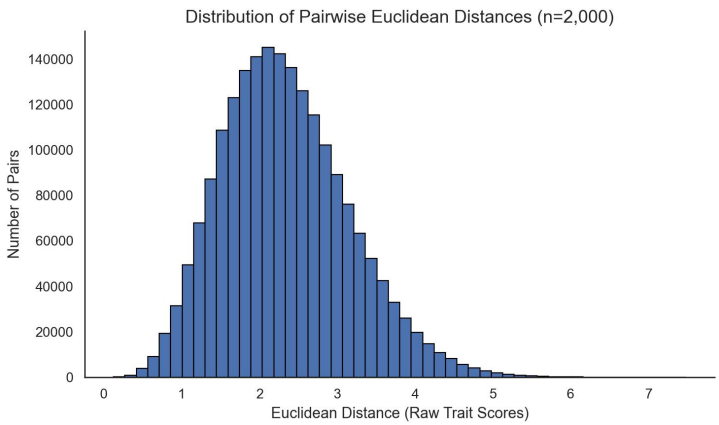


Figure 9: Compatibility or distance structure based on the Euclidean metric.

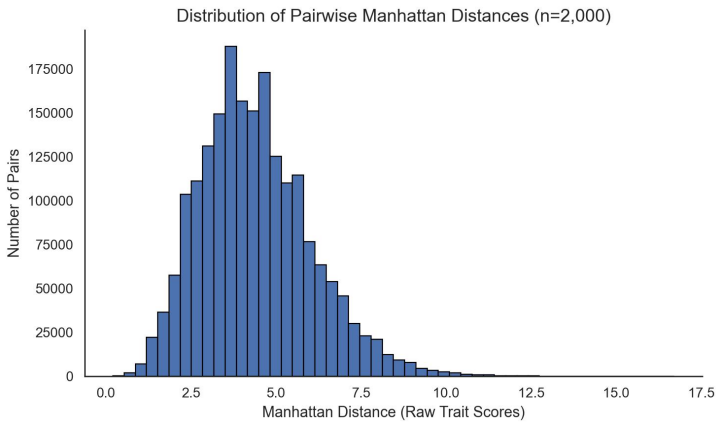


Figure 10: Compatibility or distance structure based on the Manhattan metric.

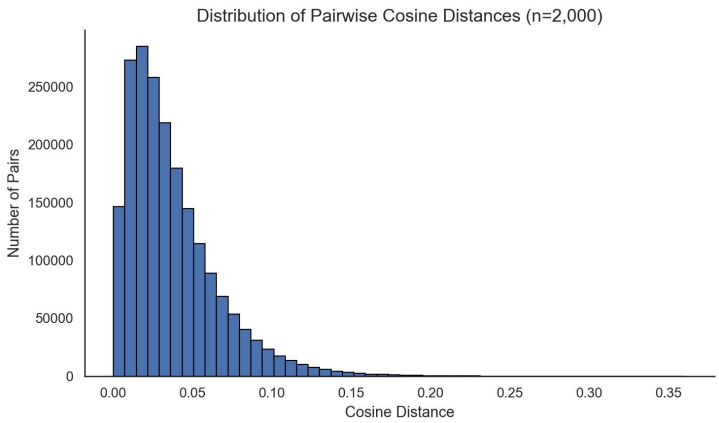


Figure 11: Compatibility or distance structure based on cosine similarity.

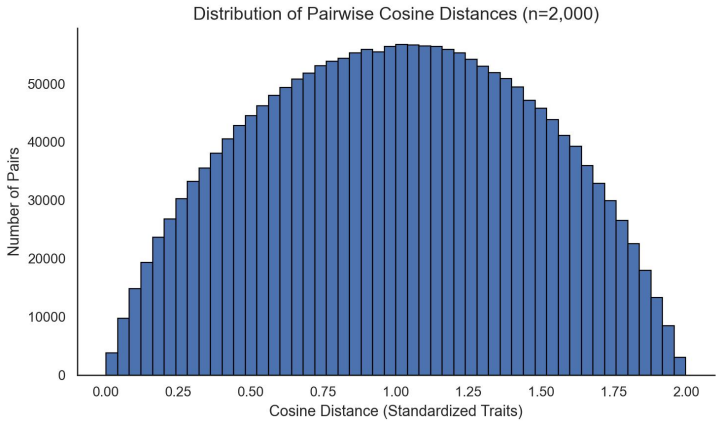


Figure 12: Compatibility or distance structure based on cosine similarity after standardisation.

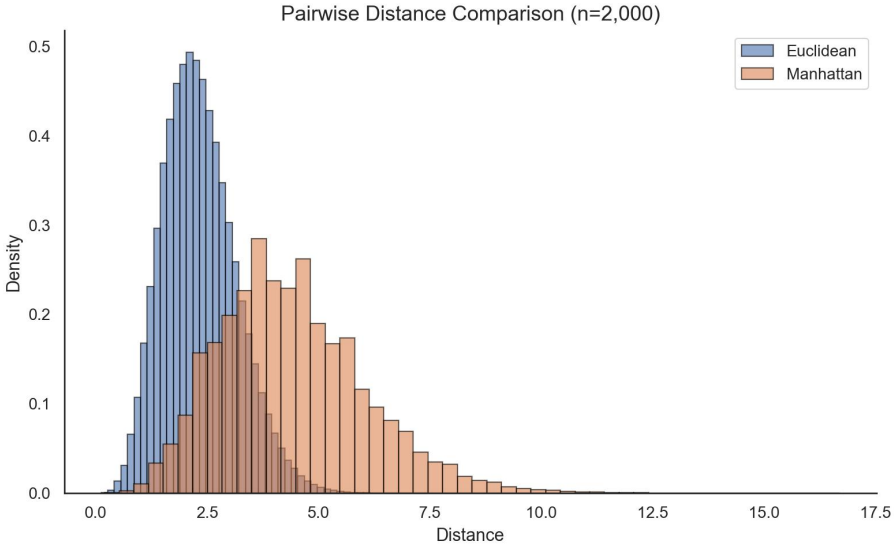


Figure 13: Comparison of compatibility patterns under Euclidean and Manhattan distance.

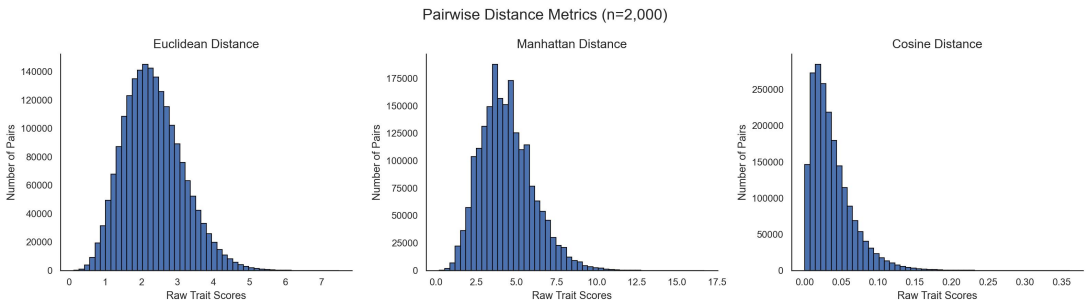


Figure 14: Comparison of pairwise results across alternative distance metrics in Stage 1.

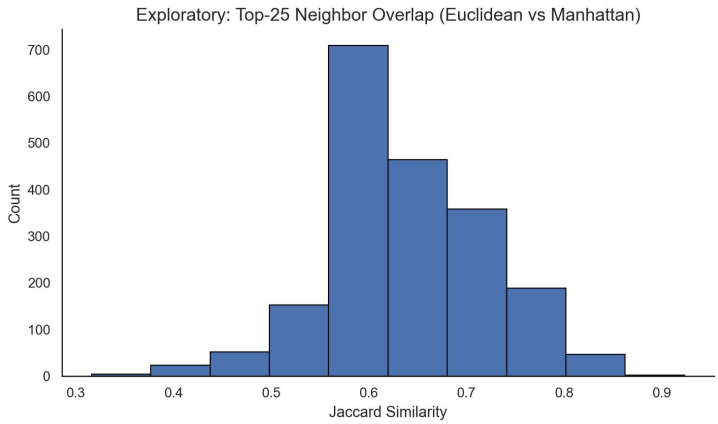


Figure 15: Jaccard overlap across alternative compatibility or ranking outputs in Stage 1.

A.3 Stage 1: Sensitivity and Assignment Comparison

These figures provide compatibility scores across weighting specifications and additional robustness checks and comparative results for the Stage 1 framework, including sensitivity analysis and a comparison between random and compatibility-guided assignment.

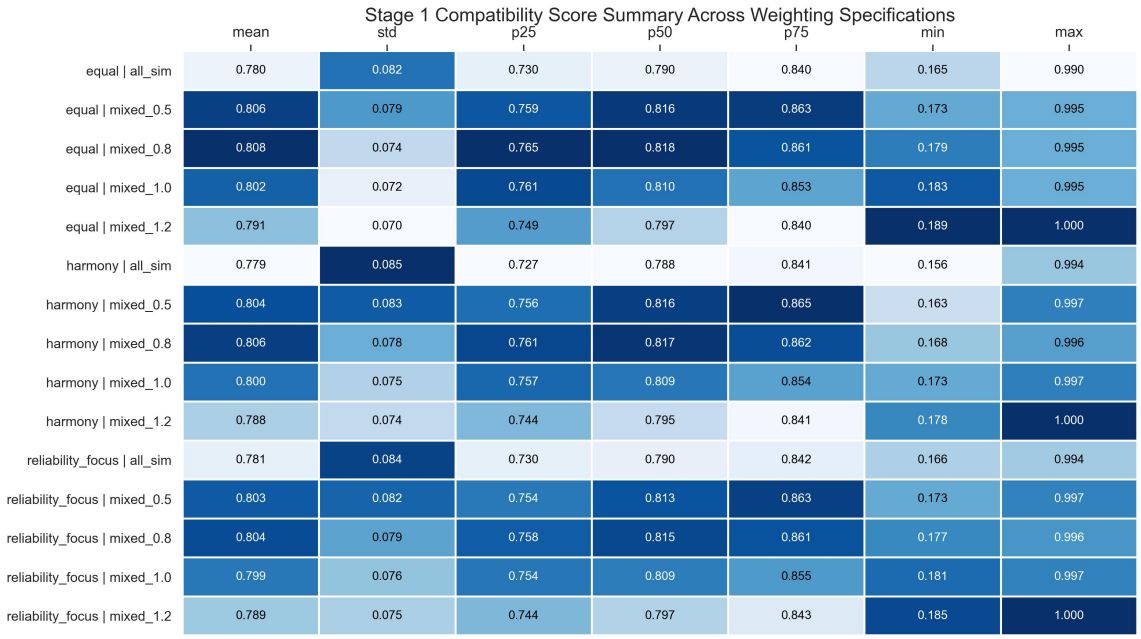


Figure 16: Heatmap summarizing compatibility patterns across the Stage 1 sample.

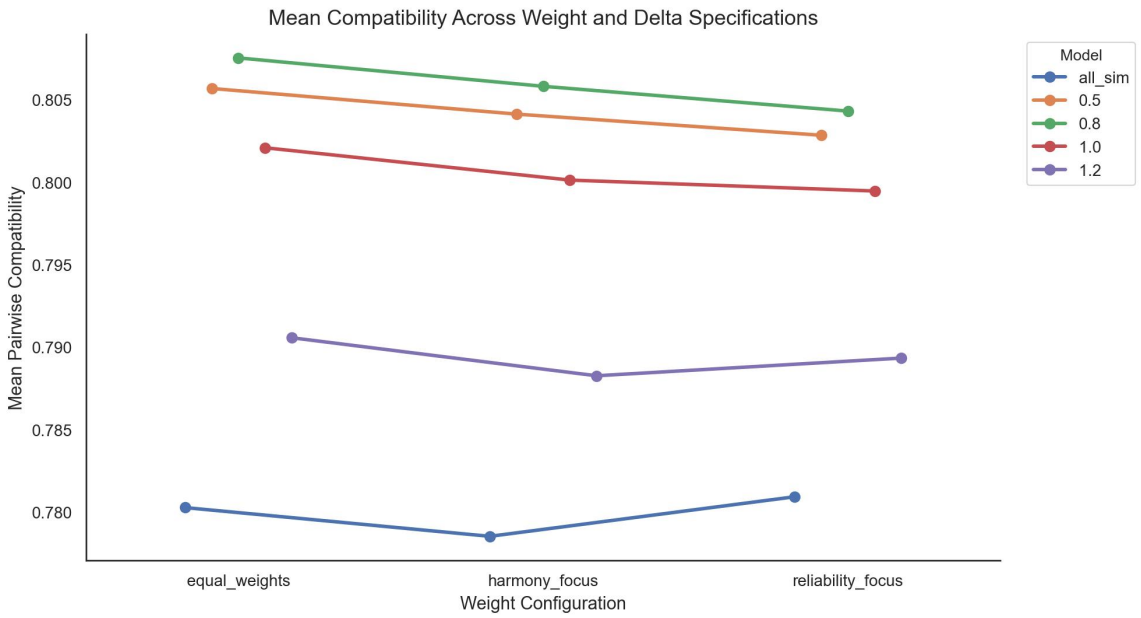


Figure 17: Sensitivity analysis for the Stage 1 compatibility framework under alternative specifications.

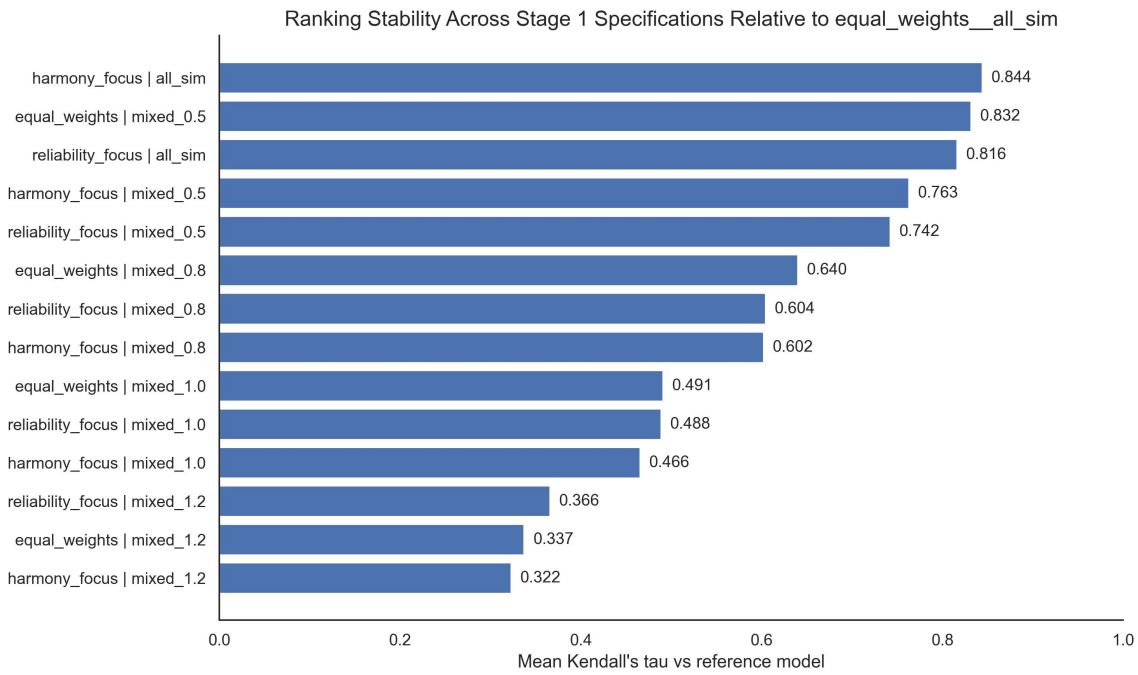


Figure 18: Rank-order sensitivity assessed using Kendall's tau in Stage 1.

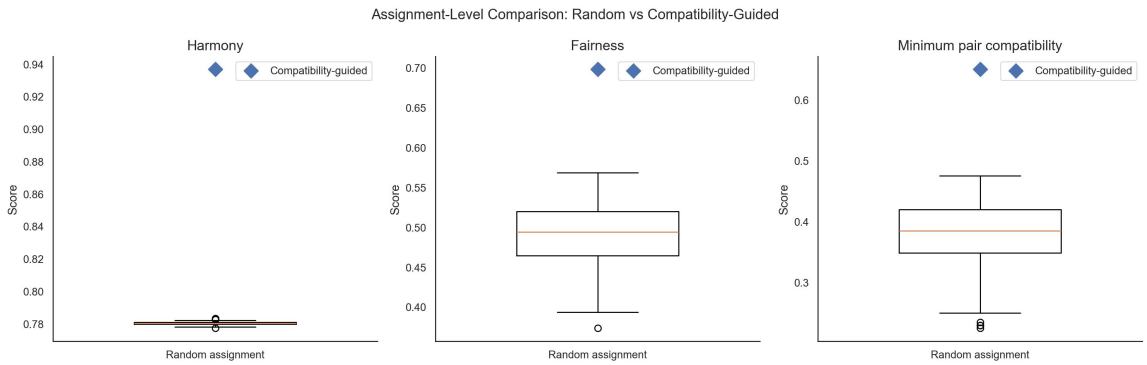


Figure 19: Comparison of assignment-level outcomes under random versus compatibility-guided allocation.

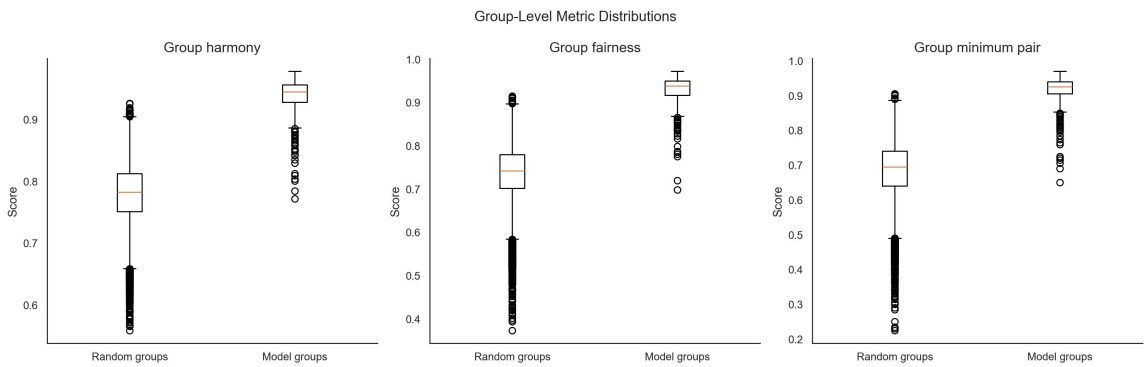


Figure 20: Comparison of assignment-level outcomes under random versus compatibility-guided allocation.

Figure 21: Comparison of assignment-level outcomes under random versus compatibility-guided allocation.

A.4 Stage 2: Descriptive Extensions

The final set of figures reports supplementary descriptive results for the extended Stage 2 framework, covering demographic, household, and lifestyle-related variables included in the broader compatibility representation.

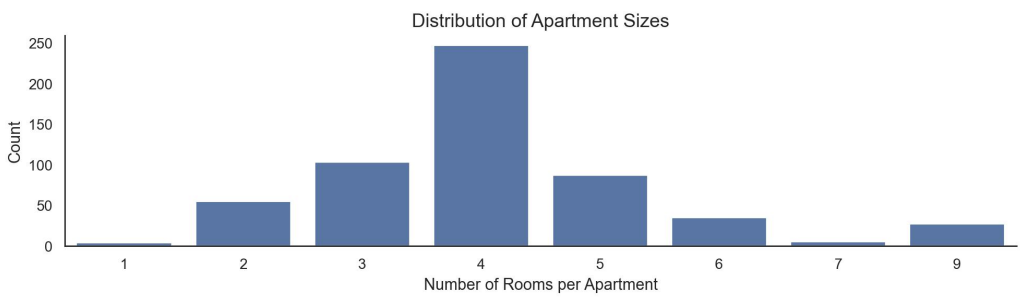


Figure 22: Distribution of apartment size in the Stage 2 dataset.

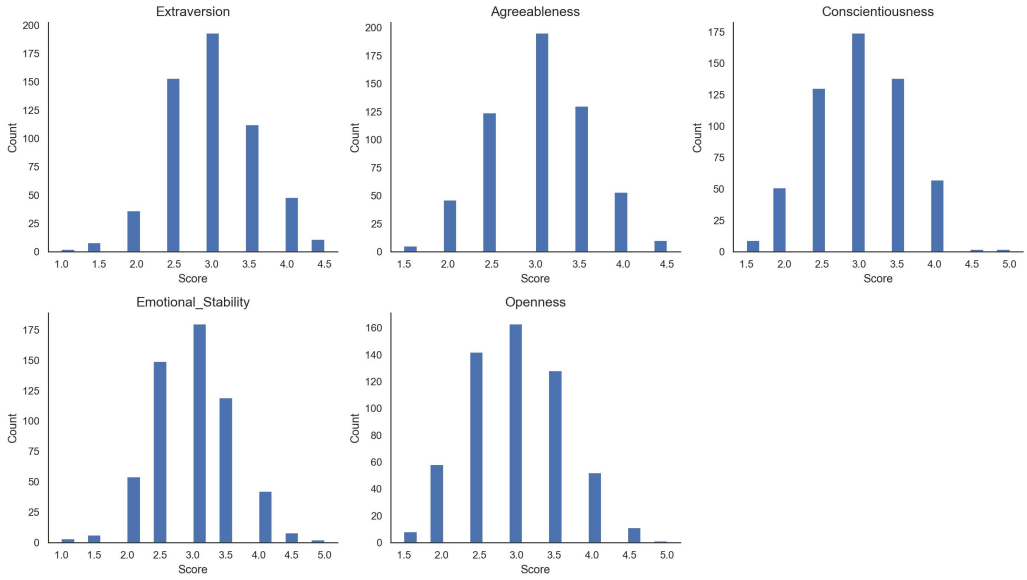


Figure 23: Distribution of Big Five trait scores in the Stage 2 dataset.

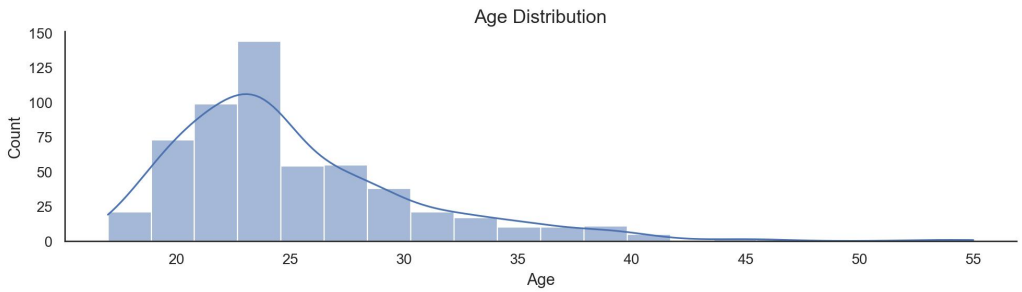


Figure 24: Age distribution in the Stage 2 dataset.

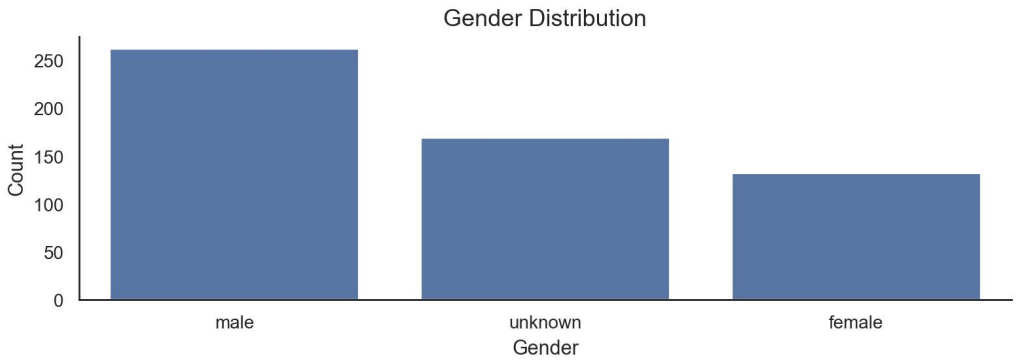


Figure 25: Gender distribution in the Stage 2 dataset.

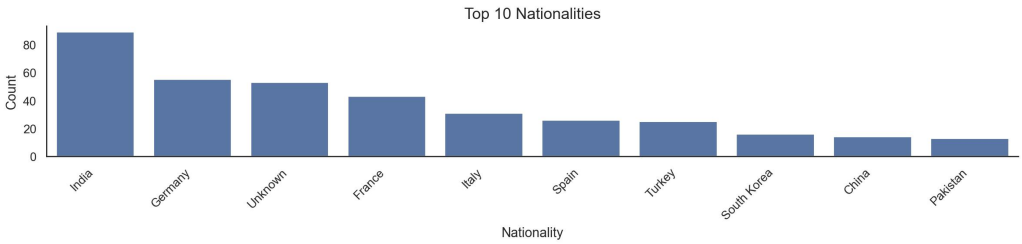


Figure 26: Nationality distribution in the Stage 2 dataset.

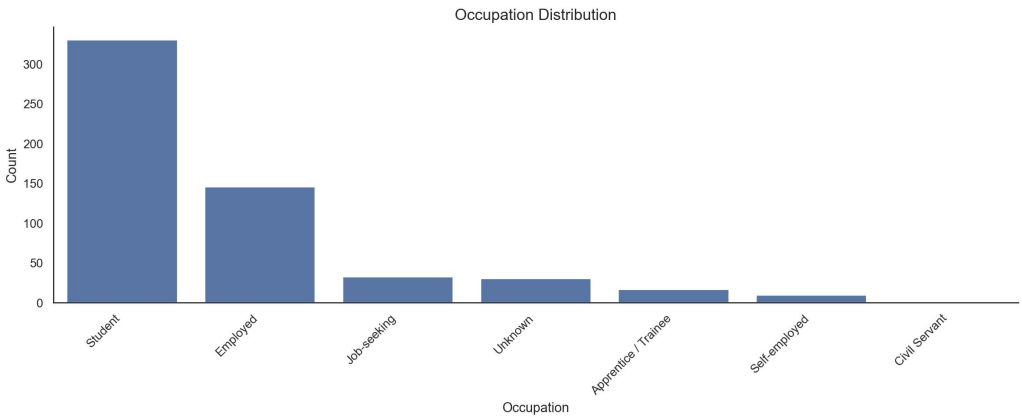


Figure 27: Occupation Distribution in the Stage-2 dataset.

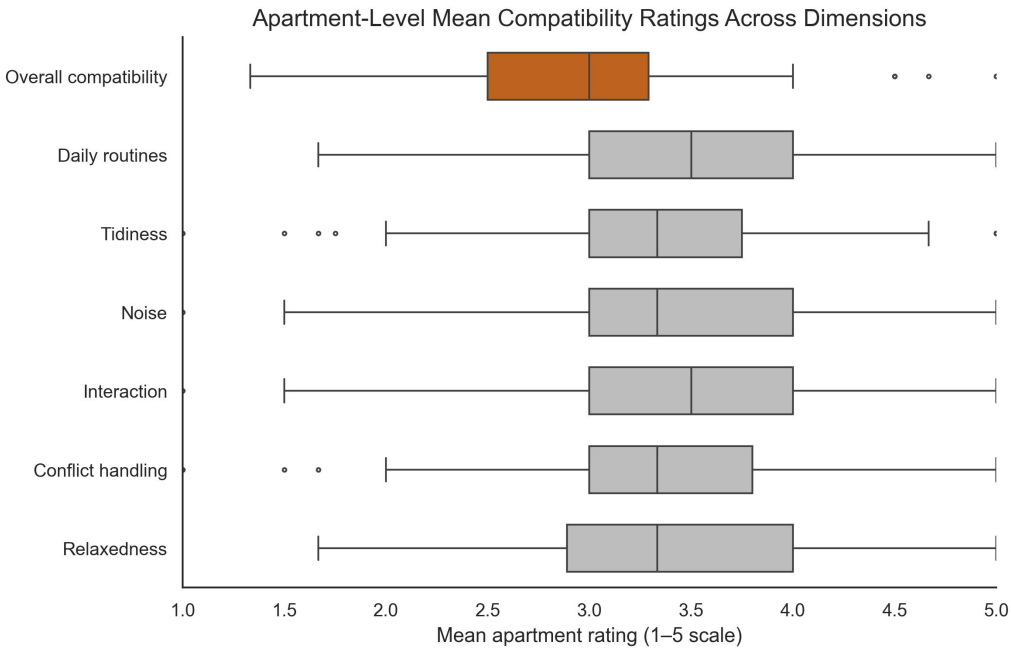
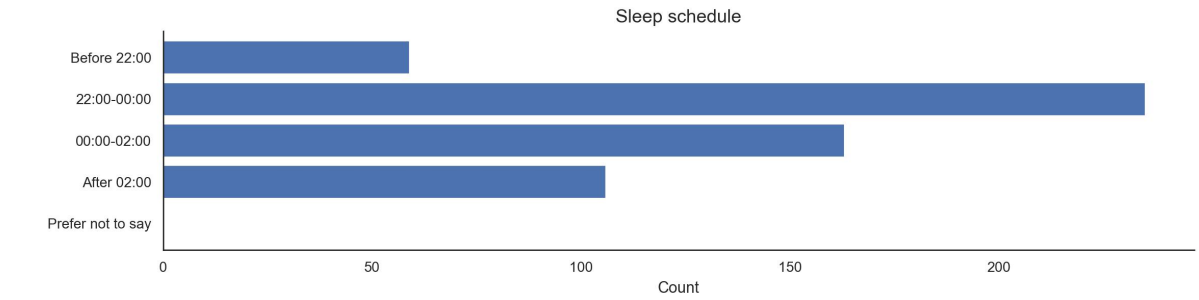
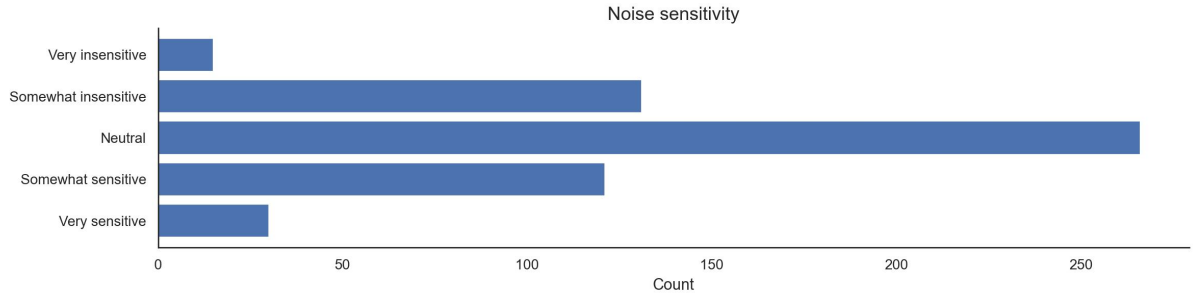
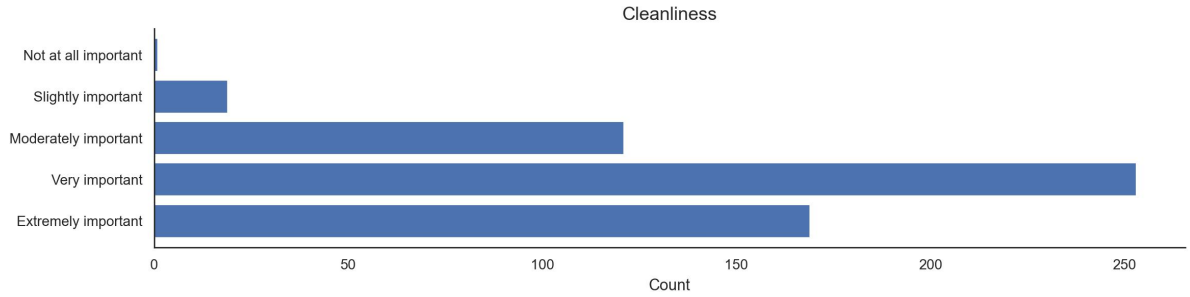
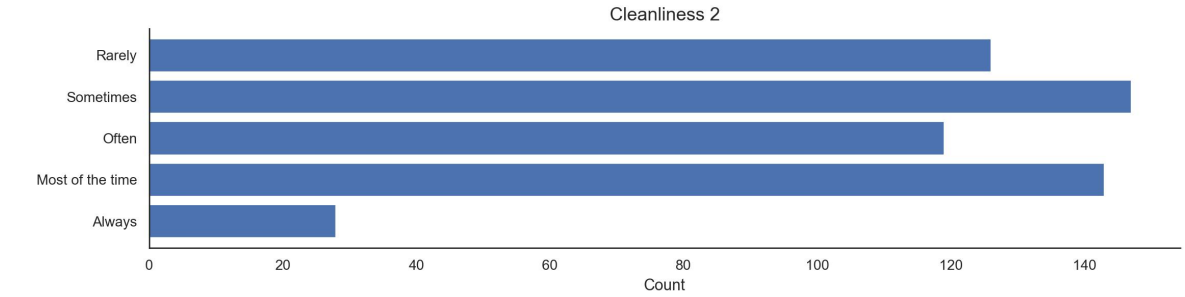


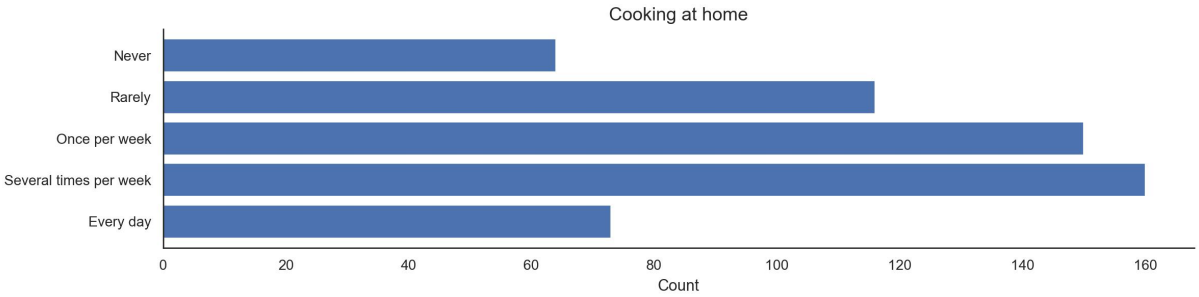
Figure 28: Boxplot of Compatibility Ratings in the Stage-2 dataset

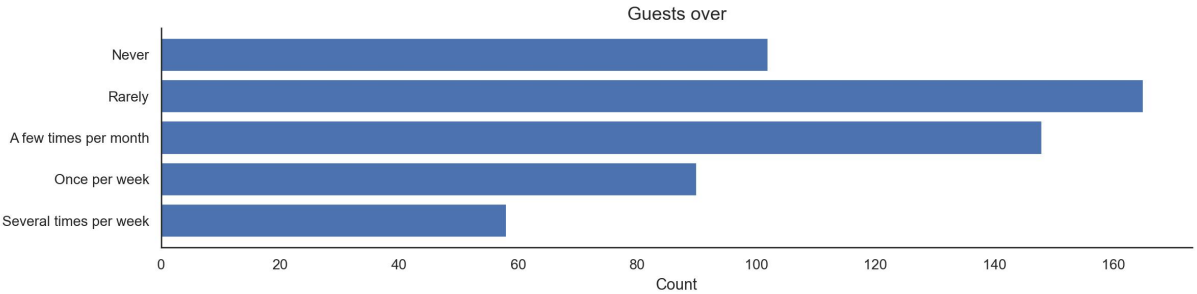

Sleep schedule
Before 22:00
22:00-00:00
00:00-02:00
After 02:00
Prefer not to say
0
50
100
150
200
Count


Noise sensitivity
Very insensitive
Somewhat insensitive
Neutral
Somewhat sensitive
Very sensitive
0
50
100
150
200
250
Count


Cleanliness
Not at all important
Slightly important
Moderately important
Very important
Extremely important
0
50
100
150
200
250
Count


Cleanliness 2
Rarely
Sometimes
Often
Most of the time
Always
0
20
40
60
80
100
120
140
Count


Cooking at home
Never
Rarely
Once per week
Several times per week
Every day
0
20
40
60
80
100
120
140
160
Count


Guests over
Never
Rarely
A few times per month
Once per week
Several times per week
0
20
40
60
80
100
120
140
160
Count

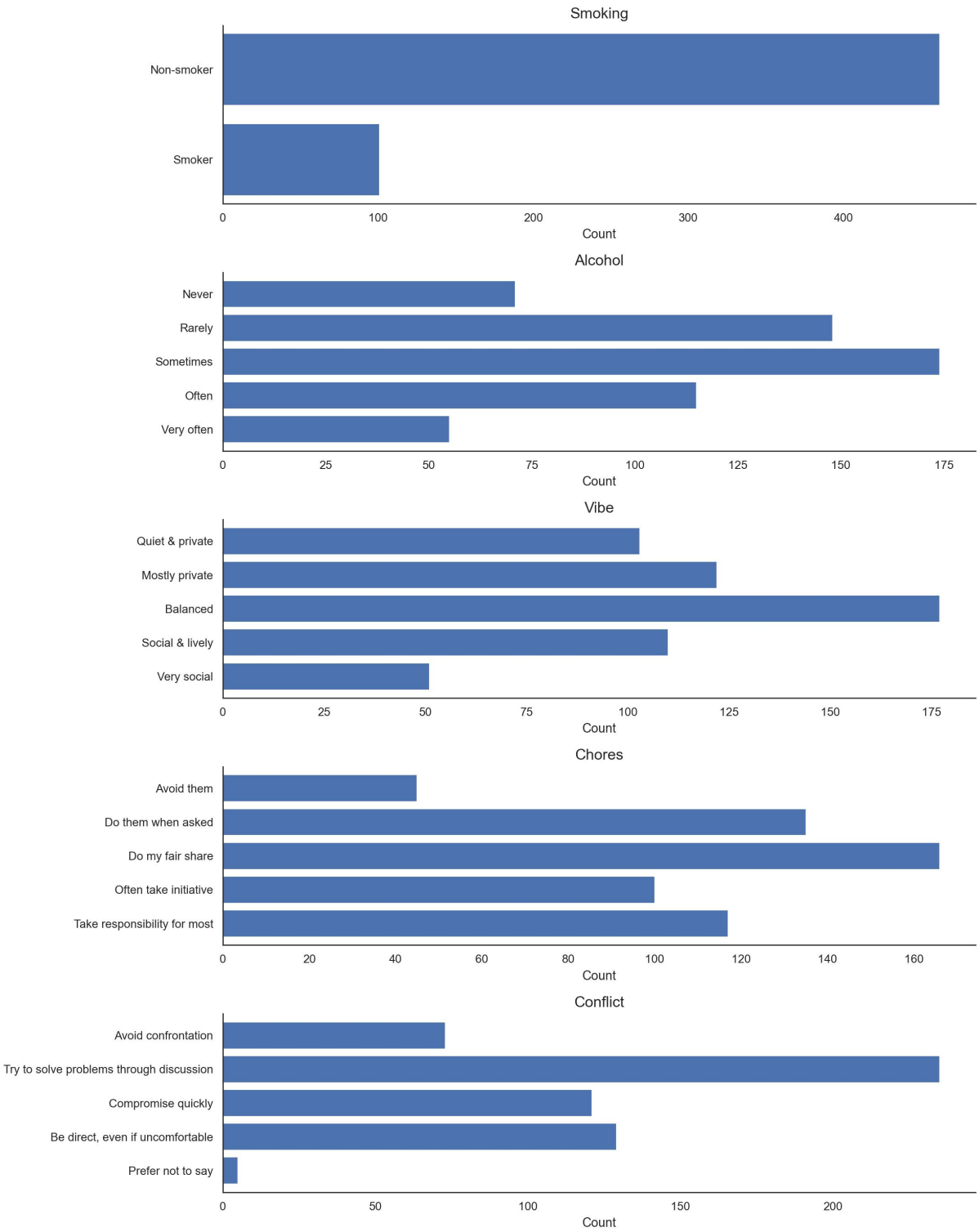


Figure 29: Distribution of selected lifestyle and behavioural pattern variables in Stage 2.

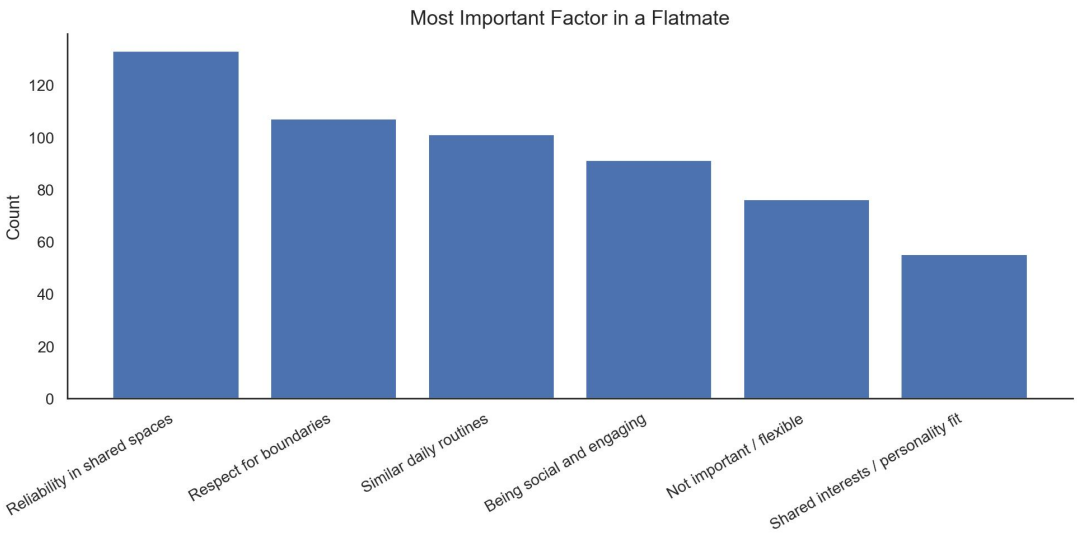


Figure 30: Distribution of reported flatmate priority in the Stage 2 dataset.

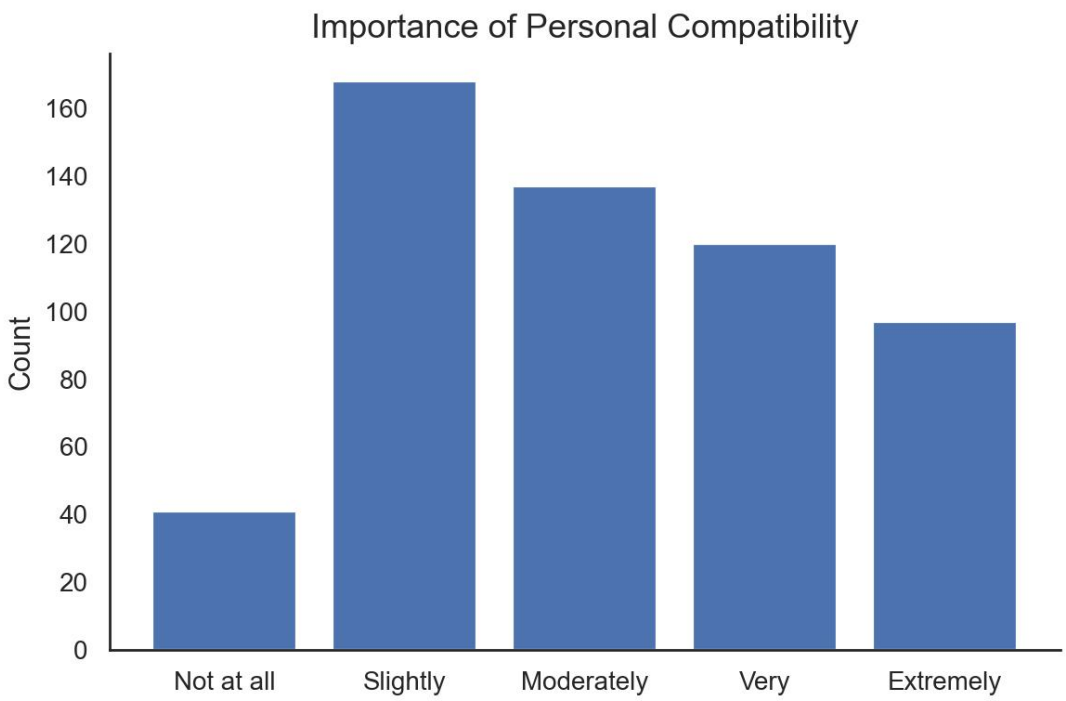


Figure 31: Reported importance of personal compatibility in the Stage 2 dataset.

A.5 Tenant Compatibility Survey

Information & Consent

What Makes a Great Flatmate Match?

Having lived at Village M myself, I'm curious about what makes flatmate groups work well. This survey explores which factors contribute to compatibility and how these might be modelled in co-living environments.

Participant Information & Consent

You are invited to take part in a research study examining compatibility and living experiences in shared co-living apartments. This study is part of a Master's thesis in Business Analytics at Católica Lisbon School of Business & Economics, conducted in collaboration with VillageM.

Purpose of the study

The goal of this research is to better understand which factors—such as lifestyle preferences, personality traits, expectations and basic demographic characteristics—contribute to harmonious co-living environments. Insights from this study may help improve future flatmate matching and co-living operations.

What participation involves

If you choose to participate:

- You will answer questions about your personality, lifestyle and routine preferences, and basic demographics.
- You will also be asked about how compatible you feel living within your co-living environment.
- The survey takes approximately 5 minutes to complete.

Voluntary participation

- Your participation is entirely voluntary.
- You can end the survey at any time simply by closing your browser.
- There are no negative consequences if you choose not to participate.

Confidentiality & data protection

- Your responses are collected anonymously.

- Your name will not be asked at any point.
- We will ask for your apartment number only to group responses by flat, not to identify individuals.
- All data will be stored securely and used solely for academic research and internal analysis.
- Results will only be reported in aggregated form; no individual will be identifiable in any publication or presentation.

Risks & benefits

- There are no anticipated risks.
- You will not receive compensation, but your input may help improve roommate matching and co-living quality in the future.

Contact

If you have questions about the study, you can contact:
Researcher: Jasper Sänger, Master of Business Analytics, Católica Lisbon School of Business & Economics, s-jsanger@ucp.pt

By clicking “I agree to participate” on the next question, you confirm that:

- You are 18 years or older
- You have read and understood the information above
- You voluntarily agree to take part in this study.

Do you consent to participate in this study under the conditions described above?

- ○ I agree to participate.
- ○ I do not agree to participate.

Building & Demographics.

What's your age?

- ○ 18–21
- ○ 22–25
- ○ 26–29
- ○ 30–33

- 34–36
- 37+

How do you identify your gender?

- Male
- Female
- Non-binary / third gender
- Prefer not to say
- [ ] Prefer to self-describe

What is your current occupation or status?

- Student
- Full-time employed
- Part-time employed
- Freelancer
- Intern / Trainee
- Currently seeking employment
- [ ] Other (please specify)

Which apartment unit do you currently reside in?

Example: A.23, HD.31, MO.71

Your response is used solely for grouping purposes. It will be removed before any analysis output to ensure anonymity.

Personality (IPIP)

Please indicate how much you agree or disagree with each statement.

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
I see myself as someone who is relaxed, handles stress well.	○	○	○	○	○

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
I see myself as someone who is outgoing, sociable.	○	○	○	○	○
I see myself as someone who has few artistic interests.	○	○	○	○	○
I see myself as someone who tends to be lazy.	○	○	○	○	○
I see myself as someone who has a forgiving nature.	○	○	○	○	○
I see myself as someone who gets nervous easily.	○	○	○	○	○
I see myself as someone who has an active imagination.	○	○	○	○	○
I see myself as someone who tends to find fault with others.	○	○	○	○	○
I see myself as someone who does a thorough job.	○	○	○	○	○
I see myself as someone who is reserved.	○	○	○	○	○

Lifestyle & Behaviour Preferences

How would you describe your typical sleep schedule?

- ○ I usually go to bed before 22:00
- ○ I usually go to bed between 22:00–00:00
- ○ I usually go to bed between 00:00–02:00
- ○ I usually go to bed after 02:00
- ○ Prefer not to say

How sensitive are you to noise at home?

- ○ Very insensitive
- ○ Somewhat insensitive

- Neutral
- Somewhat sensitive
- Very sensitive

How important is cleanliness in shared spaces (kitchen, living room, bathroom)?

- Not at all important
- Slightly important
- Moderately important
- Very important
- Extremely important

How often do you clean up immediately after using shared spaces?

- Always
- Most of the time
- Often
- Sometimes
- Rarely

How often do you cook at home?

- Every day
- Several times per week
- Once per week
- Rarely
- Never

How often do you invite guests into the apartment?

- Several times per week
- Once per week
- A few times per month
- Rarely
- Never

Do you smoke?

- Yes
- No

How often do you drink alcohol in the apartment?

- Very often
- Often
- Sometimes
- Rarely
- Never

In your daily routine at home, you prefer…

- A quiet and private environment
- Some social interaction, but mostly private time
- A balanced mix of social and private
- A social, lively environment
- Very social, frequent interaction

How would you describe your approach to shared chores (trash, dishes, common areas)?

- I avoid them
- I do them when asked
- I do my fair share
- I often take initiative
- I usually take responsibility for most chores

When conflicts arise in the flat, I tend to…

- Avoid confrontation
- Try to solve problems through discussion
- Compromise quickly
- Be direct, even if uncomfortable
- Prefer not to say

Expectations

Which factors matter most for you in a flatmate?

- [ ] Similar daily routines (sleep, noise, cleanliness)
- [ ] Respect for boundaries and personal space
- [ ] Shared interests or personality fit
- [ ] Being social and engaging
- [ ] Reliability and responsibility in shared spaces
- [ ] Not important / I am flexible
- [ ] [ ] Something else, please specify

How important is it for you to “click” on a personal level with your flatmates?

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Very
- ○ Extremely

Apartment Level Compatibility

Thinking about your apartment as a whole (all the people you share it with):
How compatible do you feel with your current flatmates overall?

- ○ Not at all compatible
- ○ Slightly compatible
- ○ Moderately compatible
- ○ Very compatible
- ○ Extremely compatible

Please indicate how much you agree or disagree with each statement about your current apartment (all flatmates together).

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
Our daily routines (sleep, work, social life) fit reasonably well together.	○	○	○	○	○
We have similar expectations about cleanliness and tidiness in shared spaces.	○	○	○	○	○
Noise levels in the apartment are at a level I can comfortably live with.	○	○	○	○	○
I feel comfortable interacting with my flatmates.	○	○	○	○	○
Conflicts or tensions in the apartment are rare or resolved quickly.	○	○	○	○	○
I feel relaxed and at ease in the shared areas of the apartment.	○	○	○	○	○

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